

Shor's Algorithm

A complete derivation, with circuits, costs, and success probabilities

Working notes

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Abstract

Let N be an odd composite and choose $a \in (\mathbb{Z}/N\mathbb{Z})^\times$. Write $r = \text{ord}_N(a)$, and let U_a be the modular-multiplication unitary $U_a|y\rangle = |ay \bmod N\rangle$, extended by the identity outside $\{0, \dots, N-1\}$. On the span of the orbit of $|1\rangle$, U_a is the cyclic shift on $\mathbb{Z}/r\mathbb{Z}$, and the Fourier matrix diagonalises that shift, so the eigenvalues of U_a there are $e^{2\pi i s/r}$. Quantum phase estimation followed by the inverse Fourier transform produces $\ell/Q \approx s/r$; continued fractions recover a divisor of r , repeated samples recover r , and $\gcd(a^{r/2} \pm 1, N)$ yields a factor for a good base. The running example is $N = 21, a = 2, r = 6$. The dominant quantum cost is reversible modular exponentiation, not the QFT.

Part I (§§1–8) is the derivation in dependence order and is the material for a talk; Part II (appendices) discharges every claim.

$$a \xrightarrow{\S 2} U_a|_{\mathcal{H}_a} \cong S_r \xrightarrow{\S 2} e^{2\pi i s/r} \xrightarrow[\S 3]{\text{QPE}} \frac{\ell}{Q} \approx \frac{s}{r} \xrightarrow[\S 5]{\text{CF}} r \xrightarrow{\S 1} \gcd(a^{r/2} \pm 1, N)$$

Notation and primitives

Let $N \geq 3$ be the integer to factor and $n := \lceil \log_2 N \rceil$. For $a \in (\mathbb{Z}/N\mathbb{Z})^\times$ the *multiplicative order* is

$$r := \text{ord}_N(a) := \min\{m \geq 1 : a^m \equiv 1 \pmod{N}\}.$$

The n -qubit *work register* has computational basis $\{|y\rangle : 0 \leq y < 2^n\}$. Define the modular-multiplication operator

$$U_a|y\rangle := \begin{cases} |ay \bmod N\rangle, & 0 \leq y < N, \\ |y\rangle, & N \leq y < 2^n. \end{cases} \quad (1)$$

The second branch is padding: it extends multiplication modulo N to a permutation of all 2^n computational-basis states, which is what makes U_a unitary. Its powers cost nothing extra,

$$U_a^m = U_{a^m \bmod N}, \quad \text{in particular} \quad U_a^{2^j} = U_{a^{2^j} \bmod N}, \quad (2)$$

because the constants $a^{2^j} \bmod N$ are computed classically by j squarings. A *controlled* unitary is

$$c\text{-}U : |c\rangle|\psi\rangle \mapsto |c\rangle U^c|\psi\rangle, \quad c \in \{0, 1\}. \quad (3)$$

The *counting register* has t qubits and we put $Q := 2^t$. With $\omega_Q := e^{2\pi i/Q}$, the Fourier matrix on Q points is

$$(F_Q)_{y,x} := \frac{1}{\sqrt{Q}} \omega_Q^{yx}, \quad 0 \leq x, y < Q, \quad \text{i.e.} \quad F_Q = \frac{1}{\sqrt{Q}} \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega_Q & \omega_Q^2 & \cdots & \omega_Q^{Q-1} \\ 1 & \omega_Q^2 & \omega_Q^4 & \cdots & \omega_Q^{2(Q-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega_Q^{Q-1} & \omega_Q^{2(Q-1)} & \cdots & \omega_Q^{(Q-1)^2} \end{pmatrix}. \quad (4)$$

Throughout, $\text{QFT}_Q := F_Q$ with the *positive* exponent, so that

$$\text{QFT}_Q|x\rangle = \frac{1}{\sqrt{Q}} \sum_{y=0}^{Q-1} e^{2\pi i xy/Q} |y\rangle, \quad \text{QFT}_Q^\dagger|x\rangle = \frac{1}{\sqrt{Q}} \sum_{y=0}^{Q-1} e^{-2\pi i xy/Q} |y\rangle. \quad (5)$$

Phase estimation applies $F_Q^\dagger$, the negative exponent, to decode a phase ramp. Fixing this once removes most sign and dagger confusion. Finally S_r denotes the cyclic shift $S_r|k\rangle = |k+1 \bmod r\rangle$ on $\mathbb{C}^r$ (the one-qubit phase gate $S = \text{diag}(1, i)$ appears only in Appendices A and C), and φ is Euler's totient.

SHORFACTOR(N) — contract

Input. A composite $N \geq 3$, $n = \lceil \log_2 N \rceil$. Even N and perfect powers are stripped classically first.

Output. A non-trivial factor d : $1 < d < N$ and $d \mid N$.

One attempt.

1. Draw $a \in \{2, \dots, N-1\}$ uniformly; if $d = \gcd(a, N) > 1$, return d .
2. $r \leftarrow \text{ORDERFIND}(N, a)$ *(the quantum subroutine; contract below)*
3. If r is even and $a^{r/2} \not\equiv -1 \pmod{N}$, return $\gcd(a^{r/2} - 1, N)$; by Lemma 1 $\gcd(a^{r/2} + 1, N)$ is then a non-trivial factor too.
4. Otherwise draw a new a and repeat.

Correctness. Step 3's hypothesis holds with probability $\geq \frac{1}{2}$ over a (Theorem 2). Every candidate is checked classically by division, so the algorithm is *Las Vegas*: a failed attempt costs a repetition and never returns a wrong factor.

Cost. $\tilde{O}(n^3)$ quantum operations on $\Theta(n)$ logical qubits, plus polynomial-time classical post-processing; expected $O(1)$ attempts.

ORDERFIND(N, a) — contract for one circuit execution

Promise. $\gcd(a, N) = 1$; $r = \text{ord}_N(a) < N$.

Available operation. Not an oracle: a reversible circuit for controlled multiplication by the *classically precomputed constant* $a^{2^j} \bmod N$, built in Appendix E. Its cost dominates the algorithm.

Parameters. $t := \lceil 2 \log_2 N \rceil$ and $Q := 2^t$, so $Q \geq N^2$ (§5 explains why).

Circuit. With F_Q as in (4),

$$|0\rangle^{\otimes t} |1\rangle \xrightarrow{H^{\otimes t}} \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle |1\rangle \xrightarrow{\prod_j c_j \cdot U_a^{2^j}} \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle |a^x \bmod N\rangle \xrightarrow{F_Q^\dagger \otimes \mathbb{1}} \text{measure } \ell.$$

The middle step uses $t = O(n)$ controlled modular multiplications by (2), not 2^t applications of U_a .

Guarantee (one execution). The counting-register statistics are those of drawing $s \in \{0, \dots, r-1\}$ uniformly and running phase estimation on the phase s/r (Theorem 37). Conditioned on s , the grid point nearest Qs/r is obtained with probability at least $4/\pi^2 = 0.405\dots$, and then

$$\left| \frac{\ell}{Q} - \frac{s}{r} \right| \leq \frac{1}{2Q} < \frac{1}{2N^2} \quad (\text{distance mod } 1).$$

Output of one execution. *Not* r . Continued fractions on ℓ/Q return the reduced denominator

$$q = \frac{r}{\gcd(s, r)} \Big| r,$$

a divisor of r , equal to r exactly when $\gcd(s, r) = 1$.

Recovery. Accumulate $R \leftarrow \text{lcm}(q_1, q_2, \dots)$ over independent executions until $a^R \equiv 1 \pmod{N}$; then $R = r$. Two executions already suffice with probability > 0.54 (Proposition 15). The test $a^R \equiv 1$ is what makes the whole procedure *Las Vegas*.

How to read this. §§1–8 justify every line of the two boxes and assume only that a quantum computer applies unitaries to a state vector and measures in the computational basis. Appendix A states the quantum mechanics used; Appendix B gives the linear algebra of the cyclic shift and its circulant/CRT background; Appendix C builds the QFT circuit; Appendix D proves the exact output distribution and the $4/\pi^2$ bound; Appendix E builds the reversible arithmetic. Nothing in §§1–8 depends on reading them.

Running example. Throughout, $N = 21$ and $a = 2$:

$$1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 11 \rightarrow 1 \pmod{21}, \quad \text{so } r = \text{ord}_{21}(2) = 6.$$

$N = 21$ rather than the usual $N = 15$ because 6 is not a power of two, so phase estimation is inexact and the continued-fraction step does real work. §6 assembles one full run.

Part I — The derivation §§1–8: the argument, in dependence order.**1 Factoring reduces to order finding**

Before any quantum mechanics: what is the quantum computer being asked for? Not a factor, but the multiplicative order of a random base. This section is entirely classical.

Lemma 1 (Non-trivial square roots of 1 split N). Let $N \geq 3$ and $x^2 \equiv 1, x \not\equiv \pm 1 \pmod{N}$. Then $\gcd(x-1, N)$ and $\gcd(x+1, N)$ are non-trivial factors of N .

Proof. $N \mid (x-1)(x+1)$. Put $d = \gcd(x-1, N)$. If $d = N$ then $x \equiv 1$, excluded. If $d = 1$ then N is coprime to $x-1$, so $N \mid x+1$, i.e. $x \equiv -1$, excluded. Hence $1 < d < N$ and $d \mid N$; apply the same to $-x$. $\square$

If $r = \text{ord}_N(a)$ is even and $a^{r/2} \not\equiv -1$, then $x := a^{r/2}$ qualifies: $x^2 = a^r \equiv 1$; $x \not\equiv 1$ else $r/2$ would already be a period, contradicting minimality.

Running example. $N = 21, a = 2, r = 6$. Then r is even and $2^3 = 8 \not\equiv -1 \equiv 20$, so the base is good, and

$$\gcd(2^3 - 1, 21) = \gcd(7, 21) = 7, \quad \gcd(2^3 + 1, 21) = \gcd(9, 21) = 3.$$

Everything from §2 onwards exists only to obtain the number 6.

Theorem 2 (A random base works). Let $N = p_1^{\alpha_1} \cdots p_m^{\alpha_m}$ be odd and composite with $m \geq 2$ distinct prime factors, and let a be uniform in $(\mathbb{Z}/N\mathbb{Z})^\times$ with $r = \text{ord}_N(a)$. Then

$$\Pr[r \text{ even and } a^{r/2} \not\equiv -1 \pmod{N}] \geq 1 - \frac{1}{2^{m-1}} \geq \frac{1}{2}. \quad (6)$$

Proof. By CRT, $(\mathbb{Z}/N\mathbb{Z})^\times \cong \prod_j (\mathbb{Z}/N_j\mathbb{Z})^\times$ with $N_j = p_j^{\alpha_j}$, and drawing a uniformly is drawing the components a_j independently and uniformly. Let $r_j = \text{ord}_{N_j}(a_j)$, so $r = \text{lcm}(r_j)$; write v_2 for the 2-adic valuation, $d_j := v_2(r_j)$, $d := v_2(r) = \max_j d_j$.

Step 1: the bad event forces all d_j equal. If r is odd, every $r_j \mid r$ is odd, so all $d_j = 0$. If r is even and $a^{r/2} \equiv -1$, then reducing mod N_j gives $a_j^{r/2} \equiv -1 \not\equiv 1$ (as N_j is odd) while $a_j^{r_j} \equiv 1$; hence $r_j \mid r$ but $r_j \nmid r/2$. Were $d_j < d$, the odd part of r_j would divide the odd part of $r/2$ (equal to that of r) and $v_2(r_j) \leq d-1 = v_2(r/2)$, so $r_j \mid r/2$ — contradiction. So $d_j = d$ for all j .

Step 2: each d_j is spread out. For odd p , $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ is cyclic of even order $n_j = \varphi(N_j)$; write $c_j = v_2(n_j) \geq 1$ and $a_j = g_j^{k_j}$ with k_j uniform mod n_j . Then $r_j = n_j / \gcd(n_j, k_j)$, so $d_j = c_j - \min(v_2(k_j), c_j)$; since $\Pr[v_2(k_j) = e] = 2^{-(e+1)}$ for $0 \leq e < c_j$, every value has $\Pr[d_j = d] \leq \frac{1}{2}$.

Step 3. Fix j_0 . By independence, $\Pr[d_1 = \cdots = d_m] = \sum_{d \geq 0} \prod_j \Pr[d_j = d] \leq \sum_{d \geq 0} \Pr[d_{j_0} = d] \cdot 2^{-(m-1)} = 2^{-(m-1)}$. $\square$

Two preconditions that are easy to omit. Both are used in the proof. N odd makes each $\varphi(N_j)$ even ($c_j \geq 1$) in Step 2 — note that $(\mathbb{Z}/2^\alpha\mathbb{Z})^\times$ fails to be cyclic for $\alpha \geq 3$ — and $m \geq 2$ supplies the factor $\frac{1}{2}$ in Step 3. Both are guaranteed by stripping factors of 2 and testing whether $N = b^c$ for $c \leq \log_2 N$ before the quantum part runs. There are two independent failure probabilities, and they should not be conflated: an unlucky base (prob. $\leq \frac{1}{2}$) and a useless measurement (§5). Each costs one repetition.

2 The order becomes an eigenphase

The quantum part computes r by turning it into the *denominator of a phase*. Nothing here is quantum mechanics: it is a statement about $r \times r$ matrices. Fix N and a with $\gcd(a, N) = 1$ and let $r = \text{ord}_N(a)$.

Proposition 3. U_a of (1) is unitary, and $U_a U_b = U_{ab \bmod N}$, whence (2).

Proof. On $\{0, \dots, N-1\}$ the map $y \mapsto ay$ is injective (a is invertible mod N), hence a bijection of a finite set; on the padding range it is the identity. So U_a permutes the computational basis and is unitary. The composition law holds on basis states, and both sides are linear. $\square$

Why the controlled-power ladder stays polynomial-size. By (2), $U_a^{2^j}$ costs a single modular multiplication by a precomputed classical constant, in place of 2^j multiplications, so a ladder of t controlled powers reaches exponent $2^t \approx N^2$ at the cost of t multiplications. Without it the ladder would have 2^t gates and there would be no algorithm. This is repeated squaring, which classical modular exponentiation uses for the same reason; it is necessary for efficiency here, and it is not by itself the source of any quantum advantage.

2.1 The orbit subspace, and the shift on it

Definition 4. The *orbit* of 1 under multiplication by a and its *span* are

$$\mathcal{O}_a := \{a^k \bmod N : 0 \leq k < r\} \subseteq (\mathbb{Z}/N\mathbb{Z})^\times, \quad \mathcal{H}_a := \text{span}\{|z\rangle : z \in \mathcal{O}_a\} \subseteq \mathbb{C}^{2^n}.$$

Proposition 5 (U_a restricted to $\mathcal{H}_a$ is a cyclic shift). $\mathcal{O}_a$ has exactly r elements, so $\dim \mathcal{H}_a = r$; U_a maps $\mathcal{H}_a$ to itself; and under the unitary identification $V|a^k \bmod N\rangle := |k\rangle$ ($k \in \mathbb{Z}/r\mathbb{Z}$),

$$V(U_a|_{\mathcal{H}_a})V^\dagger = S_r, \quad S_r|k\rangle = |k+1 \bmod r\rangle.$$

Proof. The $a^k, 0 \leq k < r$, are pairwise distinct: $a^k \equiv a^{k'}$ with $k < k' < r$ would give $a^{k'-k} \equiv 1$ with $0 < k' - k < r$, contradicting minimality of r . So V maps an orthonormal basis of $\mathcal{H}_a$ to the standard basis of $\mathbb{C}^r$ and is unitary. Since $a^r \equiv 1$, the index is well defined mod r , and $VU_aV^\dagger|k\rangle = VU_a|a^k\rangle = V|a^{k+1}\rangle = |k+1 \bmod r\rangle$. $\square$

2.2 The Fourier matrix diagonalises the shift

Let $\omega_r := e^{2\pi i/r}$ and let F_r be the Fourier matrix (4) on r points, $(F_r)_{s,k} = \omega_r^{sk}/\sqrt{r}$. A two-line calculation, done in Appendix B, gives

$$F_r S_r F_r^\dagger = \text{diag}(1, \omega_r, \omega_r^2, \dots, \omega_r^{r-1}). \quad (7)$$

So the eigenvalues of the cyclic shift are the r -th roots of unity and its eigenvectors are the Fourier waves. Transporting this back through V gives the statement the algorithm rests on.

Theorem 6 (Eigenvalues of modular multiplication on the orbit subspace). For $s = 0, \dots, r-1$ put

$$|u_s\rangle := \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k/r} |a^k \bmod N\rangle \in \mathcal{H}_a. \quad (8)$$

Then

$$U_a|u_s\rangle = e^{2\pi i s/r} |u_s\rangle, \quad s = 0, \dots, r-1.$$

The r eigenvalues $e^{2\pi i s/r}$ are distinct, so $|u_0\rangle, \dots, |u_{r-1}\rangle$ form an orthonormal eigenbasis of $\mathcal{H}_a$.

Proof. Apply U_a and shift the summation index, using $a^r \equiv 1$ so that the index is well defined mod r :

$$U_a|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k/r} |a^{k+1}\rangle = \frac{1}{\sqrt{r}} \sum_{k'=0}^{r-1} e^{-2\pi i s (k'-1)/r} |a^{k'}\rangle = e^{2\pi i s/r} |u_s\rangle.$$

The r numbers $e^{2\pi i s/r}$ are distinct because $0 \leq s < r$, so eigenvectors for different s are orthogonal; each has unit norm since $|e^{-2\pi i s k/r}| = 1$ and there are r terms. They span the r -dimensional $\mathcal{H}_a$, hence form a basis. $\square$

Where this leaves us. The period r is now encoded in the eigenvalues of an $r \times r$ cyclic-shift matrix. Every such eigenvalue has the form $e^{2\pi i \theta_s}$ with

$$\theta_s = \frac{s}{r}, \quad s = 0, \dots, r-1.$$

Order finding has become the problem of estimating one of the rational phases s/r . Quantum phase estimation (§3) produces an approximation to such a phase, and continued fractions (§5) recover its denominator.

Running example. For $N = 21, a = 2$: the orbit $\mathcal{O}_2 = \{1, 2, 4, 8, 16, 11\}$ has $r = 6$ elements, U_2 permutes it cyclically, and the eigenvalues of U_2 on $\mathcal{H}_2$ are $e^{2\pi i s/6}$ for $s = 0, \dots, 5$, i.e. the phases $\{0, \frac{1}{6}, \frac{2}{6}, \frac{3}{6}, \frac{4}{6}, \frac{5}{6}\}$. Learning any one of them with $\gcd(s, 6) = 1$ gives 6.

Why Fourier vectors appear. Modular multiplication acts as a cyclic shift on the orbit of 1, and the Fourier matrix diagonalises the cyclic-shift matrix, (7). Hence the eigenvectors are Fourier waves and the eigenvalue phases are s/r . This is the same linear-algebraic phenomenon by which the DFT diagonalises circulant matrices, i.e. every matrix commuting with the shift (Appendix B). Cryptographers meet the analogous change of basis in the NTT, where it turns convolution into coordinate-wise multiplication [25, 26]; the QFT is that same matrix applied to amplitudes.

3 Phase estimation reads the eigenphase

3.1 Kickback: making an eigenvalue observable

An eigenvalue $e^{2\pi i \phi}$ of U acting on $|u\rangle$ is a *global* phase, hence invisible. One gate makes it visible.

Lemma 7 (Kickback). Let $U|u\rangle = e^{2\pi i \phi}|u\rangle$. Then for every control amplitude pair α, β ,

$$c\text{-}U[(\alpha|0\rangle + \beta|1\rangle) \otimes |u\rangle] = (\alpha|0\rangle + \beta e^{2\pi i \phi}|1\rangle) \otimes |u\rangle. \quad (9)$$

Proof. By definition $c-U|c\rangle|\psi\rangle = |c\rangle U^c|\psi\rangle$. By linearity the left side is $\alpha|0\rangle U^0|u\rangle + \beta|1\rangle U^1|u\rangle = \alpha|0\rangle|u\rangle + \beta e^{2\pi i\phi}|1\rangle|u\rangle$; the scalar $e^{2\pi i\phi}$ commutes with everything, so it may be moved to the first tensor factor and $|u\rangle$ factored out. $\square$

How to read (9). Nothing is copied and nothing is moved: the eigenvalue was already a property of the pair $(U, |u\rangle)$. What changed is *where* it sits — as a *relative* phase between the two branches of the control, where it is observable. The target is returned unchanged, so one copy of $|u\rangle$ drives an entire ladder and is not consumed. *Incorrect inference to avoid:* “ $c-U$ projects the target onto an eigenstate.” It does not. $c-U$ is a fixed linear gate; (9) factorises only because $|u\rangle$ already was an eigenvector. If $|\psi\rangle = \sum_s c_s |u_s\rangle$ with distinct ϕ_s , then $c-U(|+\rangle \otimes |\psi\rangle) = \frac{1}{\sqrt{2}} \sum_s c_s (|0\rangle + e^{2\pi i\phi_s}|1\rangle) |u_s\rangle$ is *entangled*, which is the situation in Shor’s algorithm.

Theorem 8 (The kickback ladder). *Let $Q = 2^t$ and $U|u\rangle = e^{2\pi i\phi}|u\rangle$. Prepare t counting qubits in $|0\rangle^{\otimes t}$, apply $H^{\otimes t}$, then for $j = 1, \dots, t$ apply $c-U^{2^{t-j}}$ controlled on qubit j . The result is*

$$\bigotimes_{j=1}^t \frac{|0\rangle + e^{2\pi i 2^{t-j}\phi}|1\rangle}{\sqrt{2}} \otimes |u\rangle = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} e^{2\pi i x\phi} |x\rangle \otimes |u\rangle. \quad (10)$$

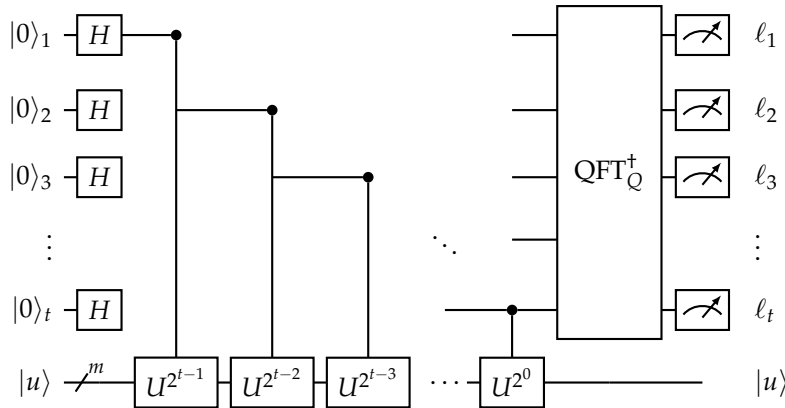
Proof. $H^{\otimes t}|0\rangle^{\otimes t} = Q^{-1/2} \sum_x |x\rangle$. Since $U^m|u\rangle = e^{2\pi i m\phi}|u\rangle$, Lemma 7 puts $e^{2\pi i 2^{t-j}\phi}$ on the $|1\rangle$ branch of qubit j and leaves the target untouched, giving the left-hand side. Expanding the product over $x = x_1 \dots x_t$ with $x = \sum_j x_j 2^{t-j}$, $\prod_j e^{2\pi i x_j 2^{t-j}\phi} = e^{2\pi i \phi \sum_j x_j 2^{t-j}} = e^{2\pi i x\phi}$, and the t independent binary sums re-label into one sum over x , each value hit once. $\square$

Qubit j carries $2^{t-j}\phi \bmod 1$; doubling shifts the binary expansion of ϕ one place left, so the top wire probes deepest into the expansion. *Measuring now teaches nothing:* every amplitude in (10) has modulus $Q^{-1/2}$, so all Q outcomes are equally likely. All the information is in relative phases, and the inverse QFT converts phase into magnitude.

3.2 The phase-estimation circuit

The transform used here is the Fourier matrix F_Q of (4), and the circuit applies its adjoint $F_Q^\dagger = \text{QFT}_Q^\dagger$, the one with the *negative* exponent (5). The reason is (7) read on Q points: F_Q turns a translation into a phase ramp, so $F_Q^\dagger$ turns a phase ramp back into a translation, i.e. into a position that a computational-basis measurement can read.

Phase estimation [3, 2] is the assembly of the three previous sections: §3 wrote the eigenphase as a ramp across a register, §C built the transform that converts a ramp into a label, and §B explains why that transform is the only one that could work. Here is the circuit and, in closed form, exactly what it returns.



Left: the Q -way beam splitter. Middle: the kickback ladder writing the ramp $e^{2\pi i x\phi}$. Right: the recombining element $\text{QFT}_Q^\dagger$ and readout. The eigenvector register is returned unchanged and can be reused. Output: $\tilde{\phi} := \ell/Q$.

Theorem 9 (State evolution). *With $U|u\rangle = e^{2\pi i\phi}|u\rangle$, the states after the three stages are*

$$|\Psi_1\rangle = \frac{1}{\sqrt{Q}} \sum_x |x\rangle |u\rangle, \quad |\Psi_2\rangle = \frac{1}{\sqrt{Q}} \sum_x e^{2\pi i x\phi} |x\rangle |u\rangle, \quad |\Psi_3\rangle = \sum_\ell \alpha_\ell |\ell\rangle |u\rangle, \quad \alpha_\ell = \frac{1}{Q} \sum_{x=0}^{Q-1} e^{2\pi i x(\phi - \ell/Q)}. \quad (11)$$

The eigenvector register never entangles with the counting register.

Proof. $|\Psi_2\rangle$ is Theorem 8. For $|\Psi_3\rangle$, apply $\text{QFT}_Q^\dagger|x\rangle = Q^{-1/2} \sum_\ell e^{-2\pi i x\ell/Q} |\ell\rangle$ to each $|x\rangle$ and collect the coefficient of $|\ell\rangle$. Factorisation of $|u\rangle$ persists because every gate acts either on the counting register alone or is a $c-U^p$ with $|u\rangle$ an eigenvector. $\square$

Theorem 10 (Exact case). If $\phi = b/Q$ for an integer b , then $|\Psi_3\rangle = |b\rangle|u\rangle$ and the output is ϕ with certainty.

Proof. $\alpha_\ell = \frac{1}{Q} \sum_x \omega_Q^{x(b-\ell)} = \delta_{b\ell}$ by Lemma 21, since $|b - \ell| < Q$. Conceptually: by (26) the state $|\Psi_2\rangle$ is $\text{QFT}_Q|b\rangle$, and $\text{QFT}_Q^\dagger$ returns $|b\rangle$ by unitarity. $\square$

3.3 What the inverse QFT does: phases become positions

F_Q diagonalises the cyclic shift, (7); $F_Q^\dagger$ therefore undoes that diagonalisation, turning eigenvalue phases back into positions. Concretely it acts as a bank of matched filters.

Concretely $\text{QFT}^\dagger$ computes, for each candidate label ℓ , the amplitude

$$\alpha_\ell = \frac{1}{Q} \sum_{x=0}^{Q-1} e^{2\pi i x \phi} e^{-2\pi i x \ell / Q} = \frac{1}{Q} \sum_{x=0}^{Q-1} e^{2\pi i x (\phi - \ell / Q)}. \quad (12)$$

Each term is a unit vector in $\mathbb{C}$; the sum is a walk of Q unit steps whose heading advances by the constant angle $2\pi(\phi - \ell/Q)$ at each step. Multiplying by $e^{-2\pi i x \ell / Q}$ *unwinds* the ramp at trial rate ℓ/Q , so α_ℓ is a test for phase matching, with three regimes:

- *Matched*, $\phi = \ell/Q$: every step points the same way, the arrows add coherently, and $\alpha_\ell = 1$ — all the probability.
- *Mismatched*, $\delta_\ell := \phi - \ell/Q \neq 0$: the headings advance by $2\pi\delta_\ell$ per step, so the Q arrows sweep a total angle $2\pi Q\delta_\ell$ and the walk curls up. Summing the geometric series,

$$\alpha_\ell = e^{\pi i (Q-1)\delta_\ell} \frac{\sin(\pi Q\delta_\ell)}{Q \sin(\pi\delta_\ell)}, \quad \text{so} \quad |\alpha_\ell| = \frac{|\sin(\pi Q\delta_\ell)|}{Q |\sin(\pi\delta_\ell)|}$$

(the leading factor is a phase and drops out of the probability; Theorem 34). The amplitude vanishes *exactly* when $Q\delta_\ell$ is a non-zero integer — the arrows are then the Q -th roots of unity taken k at a time, a balanced closed polygon, by Lemma 21 — and is $O(1/(Q|\delta_\ell|))$, hence negligible, for every ℓ more than a few grid points from $Q\phi$. The wrong answers are eliminated by this destructive interference.

- *Nearly matched*, $|\delta_\ell| \lesssim 1/(2Q)$: the arrows form a short arc instead of curling up, so the sum stays large. Only this narrow window retains appreciable amplitude; the closed form is Theorem 34.

So the inverse QFT is a bank of Q matched filters evaluated in superposition: amplitude accumulates at the label whose unwinding rate cancels the ramp, and cancels everywhere else. In the interferometer picture of §3 it is the second beam-splitter — the element that recombines the Q computational paths and erases which-path information.

Why the transform must be *inverse*. The ladder *encodes* $Q\phi$ by $|b\rangle \mapsto \text{QFT}|b\rangle$; to decode, apply $\text{QFT}^\dagger$. Using QFT instead returns $|-b \bmod Q\rangle$, since F^2 is the *negation* permutation $|k\rangle \mapsto |-k \bmod Q\rangle$ (not to be confused with the bit-reversal performed by the swap layer). That is still a correct algorithm: $\gcd(r-s, r) = \gcd(s, r)$, so the post-processing of §5 recovers the same denominator with the same probability. A *globally* flipped convention is therefore harmless, so this check will not detect one. Mixing conventions does break it: daggering the rotations while leaving the order of the $c-U^{2^{t-j}}$ alone, or omitting the swap layer on one side only. This is the usual location of the bug.

3.4 What the measurement returns

For general ϕ the sum (12) is a geometric series, and squaring it gives the exact distribution in closed form: a Fejér kernel, the intensity pattern of a Q -slit grating (Theorem 34). Three facts from it are used below, and Appendix D proves them.

The three facts needed from the output distribution. Write $\delta_\ell = \phi - \ell/Q \bmod 1$ in $[-\frac{1}{2}, \frac{1}{2})$, and let b be a grid point nearest $Q\phi$.

- Exact when it can be.* If $\phi = b/Q$ for an integer b then the outcome is b with probability 1 (Theorem 10).
- Otherwise sharply peaked.* $\Pr[b] \geq 4/\pi^2 = 0.405\dots$, and the two grid points straddling $Q\phi$ carry $\geq 8/\pi^2 = 0.811$ between them (Theorem 35). Both bounds are tight.
- Accuracy is bought with qubits.* Running with $t = \nu + \lceil \log_2(2 + \frac{1}{2\varepsilon}) \rceil$ counting qubits gives $|\ell/Q - \phi| < 2^{-\nu}$ with probability $\geq 1 - \varepsilon$ (Theorem 36). Resolution costs one qubit per bit; confidence costs $O(\log \frac{1}{\varepsilon})$ qubits, which is exponentially cheap. §5 fixes ν .

4 Why we may start from $|1\rangle$

Phase estimation as stated wants an eigenvector as input, and writing down a single $|u_s\rangle$ requires knowing r — the very quantity being computed. One line removes the circularity.

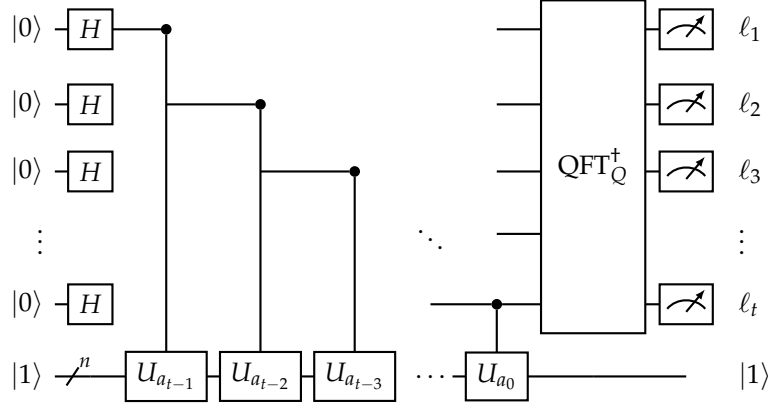
Theorem 11 (The input $|1\rangle$ is free). $\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle$.

Proof. Substituting (8) and exchanging sums, $\frac{1}{\sqrt{r}} \sum_s |u_s\rangle = \frac{1}{r} \sum_k (\sum_s e^{-2\pi i s k / r}) |a^k\rangle = \frac{1}{r} \cdot r \delta_{k,0} |a^0\rangle = |1\rangle$, by Lemma 21 (Appendix B). $\square$

This removes the apparent circularity: preparing a single $|u_s\rangle$ requires knowing r , but the *uniform superposition of all of them* is the basis state $|0 \cdots 01\rangle$, free to prepare with one X gate.

4.1 The order-finding circuit

Combining §§2–3.2: run QPE on $U = U_a$ with input $|1\rangle$. By Theorem 11 $|1\rangle = r^{-1/2} \sum_s |u_s\rangle$, and by Theorem 37 (App. D) the outcome is distributed as if s were drawn uniformly from $\{0, \dots, r-1\}$ and QPE run on $|u_s\rangle$, whose eigenphase is $\phi_s = s/r$. The work register is never measured.



$a_\ell := a^{2^\ell} \bmod N$, computed classically by ℓ squarings; by Proposition 3 each box is *one* in-place modular multiplication by a compile-time constant. The top (most significant) wire drives $U_a^{2^{t-1}}$, the bottom drives U_a .

Theorem 12 (Two descriptions of one state). *After the ladder and before $\text{QFT}^\dagger$, the joint state is*

$$\underbrace{\frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle \otimes |a^x \bmod N\rangle}_{\text{periodic-function form}} = \underbrace{\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} \left(\frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} e^{2\pi i x s / r} |x\rangle \right) \otimes |u_s\rangle}_{\text{eigenphase form}}. \quad (13)$$

Proof. Inverting (8) by Lemma 21 gives $|a^x \bmod N\rangle = r^{-1/2} \sum_s e^{2\pi i x s / r} |u_s\rangle$; substitute and exchange the sums. $\square$

Being literally the same vector, both give identical counting-register statistics. The left form is Shor’s 1994 “period finding”, the right the modern phase-estimation reading; measuring the work register and *discarding* the outcome gives the same marginal distribution on the counting register as not measuring it at all; conditioning on a particular work-register outcome does change the counting-register state. Period finding makes clear *why the modular exponentiation is there*; phase estimation makes clear *why the inverse QFT is there and why $t = 2n + O(1)$* .

5 From ℓ to r , and why $t = 2n + O(1)$

Measurement returns an ℓ for which ℓ/Q only approximates s/r . Continued fractions recover the exact rational. Recall the convergents p_k/q_k of x , generated by $p_k = a_k p_{k-1} + p_{k-2}$, $q_k = a_k q_{k-1} + q_{k-2}$ from the partial quotients a_k ; $p_k q_{k-1} - p_{k-1} q_k = (-1)^{k-1}$, so every convergent is automatically in lowest terms, and $q_k \geq F_{k+1}$ grows exponentially, so there are $O(\log q)$ of them and the whole expansion costs $O(\log^3 Q)$ bit operations.

Theorem 13 (Legendre). *If p/q is in lowest terms, $q \geq 1$, and $|x - \frac{p}{q}| < \frac{1}{2q^2}$, then p/q is a convergent of the continued-fraction expansion of x .*

Corollary 14 (Why $t \geq 2n$). *Suppose $r < N$ and $|\ell/Q - s/r| < \frac{1}{2N^2}$. Then the last convergent of ℓ/Q with denominator $< N$ is s'/q , the fraction s/r in lowest terms; so $q \mid r$, with $q = r$ iff $\gcd(s, r) = 1$. In particular $Q \geq N^2$, i.e. $t \geq 2n$, suffices.*

Proof. Write $s/r = s'/q$ in lowest terms, $q \leq r < N$. Then $|\ell/Q - s'/q| < \frac{1}{2N^2} \leq \frac{1}{2q^2}$, so Theorem 13 applies and $s'/q = p_k/q_k$ for some k . It remains to see no *later* convergent also has denominator $< N$: the standard bound $|x - p_k/q_k| \geq \frac{1}{q_k(q_{k+1} + q_k)}$ would, if $q_{k+1} < N$, force $|x - s'/q| > \frac{1}{2N^2}$, contradicting the hypothesis. $\square$

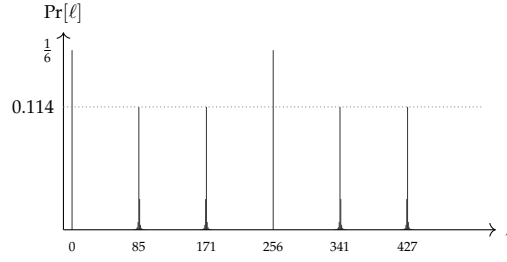


Figure 1: The exact measurement distribution for $N = 21$, $a = 2$ ($r = 6$), $t = 9$, $Q = 512$: six peaks at $\ell \approx 512s/6$, i.e. $\{0, 85, 171, 256, 341, 427\}$, carrying 78.9% of the mass. The branches $s = 0, 3$ sit on exact grid points ($512s/6 \in \mathbb{Z}$) and give $\text{Pr} = 0.16667$ each; the other four spread, with 0.114 on the nearest grid point. The peak heights are only *approximately* $1/6$ because $6 \nmid 512$: $512 = 6 \cdot 85 + 2$, so two residue classes of x hold 86 values and four hold 85, and the branch weights are perturbed at the 5×10^{-6} level. Exact numerics, not a sketch of the envelope.

Why the counting register is twice the size of the work register. The denominator being sought is as large as $r < N$, and Legendre's threshold scales as the *square* of the denominator; hence precision better than $1/(2N^2)$, hence $Q \geq N^2$, hence $t = 2n + O(1)$ — and hence $2n$ modular multiplications in place of n . The constant $\frac{1}{2}$ in Theorem 13 cannot be improved to 1 (take $x = 0.4$, $p/q = 1/3$), so the bound is tight.

Two choices of t . The algorithm box on page 1 fixes the first; the second is a remark. These are often conflated in the literature. $t = \lceil 2 \log_2 N \rceil$ (minimal; nearest grid point is within $1/(2Q) \leq 1/(2N^2)$, obtained with probability $\geq 4/\pi^2$ per branch), or $t = 2n + 1 + \lceil \log_2(2 + \frac{1}{2\epsilon}) \rceil$ (Theorem 36 with $v = 2n + 1$; accuracy $2^{-(2n+1)} \leq 1/(2N^2)$ with probability $\geq 1 - \epsilon$). Both give $t = 2n + O(1)$.

Proposition 15 (What a single run actually returns). *Condition on the outcome being accurate enough for Corollary 14 (probability $\geq 1 - \epsilon$). Then s is uniform in $\{0, \dots, r - 1\}$ and the convergent is in lowest terms, so what is recovered is $q = r / \gcd(s, r)$, a divisor of r . It equals r exactly when $\gcd(s, r) = 1$, with probability*

$$\frac{\varphi(r)}{r} = \prod_{p|r} \left(1 - \frac{1}{p}\right) = \Omega\left(\frac{1}{\log \log r}\right). \quad (14)$$

What one execution returns is therefore a divisor q of r , and the repair is to take more samples: with $q_1, q_2, \dots$ from independent runs, accumulate $R_m := \text{lcm}(q_1, \dots, q_m)$ and stop when $a^{R_m} \equiv 1 \pmod{N}$, at which point $R_m = r$. Two runs already suffice whenever $\gcd(s_1, s_2) = 1$, which has probability $\geq 1 - \sum_p p^{-2} = 1 - 0.4522 \dots > 0.54$. (Testing small multiples $2q, 3q, \dots$ of a single q is a useful practical shortcut, but it is a bounded heuristic, not a worst-case repair: $\lfloor N/q \rfloor$ can be exponential in $n = \log_2 N$.) The outcome $\ell = 0$ carries no information.

Proof. For the lcm claim: let $p^e \parallel r$. If $p^e \nmid r_1$ and $p^e \nmid r_2$ then p divides both $\gcd(s_1, r)$ and $\gcd(s_2, r)$, hence $p \mid \gcd(s_1, s_2) = 1$, absurd; so p^e divides one of them, and both divide r . The probability bound is a union bound over primes with $\text{Pr}[p \mid s_i] \approx 1/p$. $\square$

Because the recovery step *verifies* $a^R \equiv 1$ and the factoring step verifies the factor by division, a wrong order is never used. A high per-run success probability is therefore unnecessary, and noise on real hardware wastes runs; it cannot produce a wrong factorisation.

6 A complete run: $N = 21$, $a = 2$

$1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 11 \rightarrow 1 \pmod{21}$, so $r = \text{ord}_{21}(2) = 6$, even, with $2^3 = 8 \not\equiv -1 \equiv 20$: the base is good. Minimal register: $n = 5$, and $Q \geq N^2 = 441$ forces $Q = 512$, $t = 9$. The exact distribution (Figure 1) has six peaks at $\ell \in \{0, 85, 171, 256, 341, 427\} \approx 512s/6$ carrying 78.9% of the mass; branches $s = 0, 3$ sit on exact grid points ($512s/6 \in \mathbb{Z}$, $\text{Pr} = 0.16667$ each), the other four spread with 0.114 on the nearest grid point — 0.684 within their branch, above the $4/\pi^2$ floor. Post-processing:

$$\frac{85}{512} = [0; 6, 42, 2] \Rightarrow \text{convergents } 0, \frac{1}{6}, \dots \Rightarrow q = 6 \checkmark; \quad \frac{171}{512} \Rightarrow \frac{1}{3} \Rightarrow q = 3, 2^3 \neq 1;$$

$\frac{256}{512} = \frac{1}{2} \Rightarrow q = 2$; $\frac{341}{512} \rightarrow \frac{2}{3} \Rightarrow q = 3$; $\frac{427}{512} \rightarrow \frac{5}{6} \Rightarrow q = 6$; and $\ell = 0$ is uninformative. Two runs giving $q = 2$ and $q = 3$ have $\text{lcm} = 6$, and $2^6 \equiv 1$ verifies it. Exactly: a single run returns $r = 6$ *directly* with probability 0.3208 (theory: $\varphi(6)/6 = 1/3$, minus leakage), and 0.8149 after testing small multiples. Then $\gcd(2^3 - 1, 21) = \gcd(7, 21) = 7$ and $\gcd(2^3 + 1, 21) = \gcd(9, 21) = 3$.

Why not $N = 15$? $(\mathbb{Z}/15\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$, so $\text{ord}_{15}(a) \in \{1, 2, 4\}$ for every base: the period is always a power of two, phase estimation is exact, and continued fractions are not exercised at all. Nearly every published hardware demonstration nevertheless uses $N = 15$.

7 Cost and success probability

Resource figures in this section and standardisation status in §8 were last checked in August 2026; both are date-sensitive.

Theorem 16 (Resources). *For an n -bit modulus and $t = 2n + O(1)$:*

Stage	Cost	Note
$H^{\otimes t}$	$t = 2n + O(1)$ Clifford gates	negligible
modular exponentiation	$\Theta(n^3)$ Toffoli; $\approx 0.3 n^3$ windowed	$2n$ multiplications $\times n$ additions $\times \Theta(n)$
QFT_Q^+	$\Theta(n^2)$ exact / $O(n \log n)$ approximate	Thm. 33, App. C
measurements	t	or t rounds, one qubit
continued fractions	$O(n^3)$ bit operations	classical
total	$\Theta(n)$ logical qubits, $\Theta(n^3)$ Toffoli	$T \approx 4 \times$ Toffoli count

Toffoli-depth is $\Theta(n^3)$ with ripple-carry adders, since the $2n$ modular multiplications are inherently sequential; carry-lookahead adders reduce each addition to depth $O(\log n)$, giving $\Theta(n^2 \log n)$ depth at the price of $O(n)$ extra ancillas and a constant-factor worse Toffoli count.

The cost of Shor’s algorithm is arithmetic. At $n = 2048$: modular exponentiation $\approx 2.6 \times 10^9$ Toffoli against $\approx 9 \times 10^4$ gates for the approximate QFT on $t = 4097$ qubits — a ratio $\approx 3 \times 10^4$, so in this estimate the QFT is $\approx 0.003\%$ of the circuit. The percentage is specific to the construction and the parameters; the general claim, which holds asymptotically ($\Theta(n^3)$ against $O(n \log n)$) and in every published estimate, is that reversible modular arithmetic dominates. The binding cost is multiplying two 2048-bit numbers reversibly, in superposition.

Concrete estimates for RSA-2048. With windowed arithmetic, Gidney–Ekerå [15] give $\approx 3 \times 10^9$ Toffoli gates on a bit more than $3n \approx 6200$ logical qubits; Gidney 2025 [16] reaches ≈ 1400 logical qubits at 6.5×10^9 Toffoli per run with ≈ 9 runs, i.e. under a million physical qubits and about five days at a 10^{-3} gate error rate. Appendix G tabulates these against the minimal-width and low-qubit constructions and lists the variants (Ekerå–Håstad, Regev, Ragavan–Vaikuntanathan) that move the numbers. A gate count is interpretable only alongside the construction it came from.

Success probability, assembled.

Event	Probability	
base a is good (r even, $a^{r/2} \neq -1$)	$\geq 1 - 2^{-(m-1)} \geq \frac{1}{2}$	Thm. 2
QPE returns the nearest grid point, per branch	$\geq 4/\pi^2 = 0.4053$	Thm. 35
... or one of the two straddling points	$\geq 8/\pi^2 = 0.8105$	
$ \ell/Q - s/r < 2^{-(2n+1)}$ with the padded register	$\geq 1 - \varepsilon$	Thm. 36
convergent equals r outright, i.e. $\gcd(s, r) = 1$	$\varphi(r)/r = \Omega(1/\log \log N)$	Prop. 15
two runs give r via lcm	> 0.54	
per attempt (base + one run)	$\geq \frac{1}{2}(1 - \varepsilon)\varphi(r)/r = \Omega(1/\log \log N); \Omega(1)$ with the lcm variant	
overall	Las Vegas: expected $O(\log \log N)$ attempts, $O(1)$ with lcm; never a wrong factor	

Corollary 17 (Shor [1]). *There is a randomised algorithm outputting a non-trivial factor of a composite n -bit N using $O(n)$ logical qubits and $\tilde{O}(n^3)$ gates, with the answer verifiable classically.*

8 What the mechanism reaches

Shor’s algorithm breaks RSA, finite-field Diffie–Hellman, DSA, ECDH and ECDSA. All are instances of the *abelian* hidden subgroup problem, and the machinery above requires the group to be abelian: the irreducible representations of an abelian group are one-dimensional, so each eigenvalue is a single complex phase, and Lemma 7 can move it onto a control qubit. A non-abelian group has irreducible representations of dimension > 1 (alongside any one-dimensional ones: S_3 has the trivial and sign characters as well as a 2-dimensional irrep), so Fourier sampling returns matrix-valued blocks and the scalar-character argument above no longer suffices. The obstruction is not merely that some eigenvalue fails to be a phase; it is that extracting the hidden subgroup from those blocks requires measurements nobody knows how to implement efficiently in general. No efficient quantum algorithm is known for the non-abelian HSP, which is why graph isomorphism (symmetric group) remains open, and why certain lattice problems resist this attack — specifically, unique-SVP reduces to the *dihedral* HSP [19], which is a statement about that reduction and not a blanket claim about the lattice assumptions used in cryptography. The deployed post-quantum standards rest on lattice and hash assumptions: NIST finalised ML-KEM, ML-DSA and SLH-DSA as FIPS 203–205 in August 2024 [25, 26, 27], selected the code-based HQC in March 2025 [28], and has FN-DSA in preparation as FIPS 206; no isogeny-based scheme is a NIST standard. (Standardisation status last checked August 2026; it is date-sensitive.) Shor’s algorithm does not generically break symmetric primitives. Generic quantum attacks give different speed-ups for key search, preimage search and collision finding, by amplitude amplification rather than eigenvalue estimation, so

the parameter adjustment depends on the primitive and the security notion; there is no single “double every length” rule.

On timelines. No cryptographically relevant modulus has been factored by Shor’s algorithm and none is close: the gap between current demonstrations and RSA-2048 — which needs $\sim 10^9$ Toffoli gates and, with error correction, on the order of 10^6 physical qubits — is many orders of magnitude in both dimensions. Yet anything encrypted today that must stay secret for a decade should already be migrating, because an adversary can record ciphertext now and decrypt it when the machine arrives. The planning question is how long this data must stay secret, and how long migration will take.

Appendix A The quantum mechanics used

Four postulates and one consequence suffice. All spaces are finite-dimensional over $\mathbb{C}$, so completeness is automatic, operators are matrices, and every normal operator is diagonalisable.

A.1 The four postulates

- (P1) **State.** The state of an isolated system is a unit vector $|\psi\rangle$ in a Hilbert space $\mathcal{H}$, determined only up to a global phase: $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ are physically identical. One qubit is $\mathcal{H} = \mathbb{C}^2$ with distinguished orthonormal basis $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$; a k -qubit register is $(\mathbb{C}^2)^{\otimes k} \cong \mathbb{C}^{2^k}$ with basis $|x\rangle = |x_1\rangle \otimes \cdots \otimes |x_k\rangle$, and we read $x = \sum_{s=1}^k x_s 2^{k-s}$ so that a register *stores an integer* (big-endian: x_1 is the most significant bit and the top wire).
- (P2) **Evolution.** A closed system evolves by a unitary: $|\psi\rangle \mapsto U|\psi\rangle$ with $U^\dagger U = U U^\dagger = \mathbb{1}$. Gates are unitary, hence invertible: *every quantum computation is reversible*, and any classical subroutine must be rewritten reversibly before it can be used (§E).
- (P3) **Measurement (Born rule).** Measuring $|\psi\rangle$ in the computational basis returns outcome x with probability $\Pr[x] = |\langle x|\psi\rangle|^2$, and leaves the state at $|x\rangle$. If only a subregister A is measured, $\Pr[x] = \|(|x\rangle\langle x| \otimes \mathbb{1})|\psi\rangle\|^2$ and the state collapses to the normalised projection $(|x\rangle\langle x| \otimes \mathbb{1})|\psi\rangle / \sqrt{\Pr[x]}$.
- (P4) **Composition.** Systems compose by tensor product: $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$. So $t + n$ qubits carry 2^{t+n} complex amplitudes — the exponential index space the algorithm ranges over.

The consequence:

(P5) **Interference.** Amplitudes add *before* they are squared. If outcome x is reachable by several computational paths with amplitudes $A_1, A_2, \dots$, then $\Pr[x] = |\sum_j A_j|^2$; the classical value would be $\sum_j |A_j|^2$. The cross terms $2 \operatorname{Re}(A_j \overline{A_{j'}})$ are what let wrong answers cancel.

The price. Suppose some register — an ancilla, a scratch bit, an uncleared carry — is left holding a record $|g_j\rangle$ of which path was taken. Tracing it out, the cross term for paths $j \neq j'$ acquires the factor $\langle g_{j'}|g_j\rangle$: it vanishes *exactly* when the records are orthogonal (perfect which-path information) and is suppressed in proportion to the overlap otherwise, so $\Pr[x]$ degrades towards the classical $\sum_j |A_j|^2$. What matters is therefore not that scratch space is never written, but that branch-*dependent* records are uncomputed — decoupled from the branch label — before the branches they distinguish are recombined. Every arithmetic block in §E accordingly restores its ancillas to a state independent of the branch, by convention $|0\rangle$, which also makes them reusable. Failing to do so is the commonest source of silent bugs.

What is *not* true: superposition does not “try all answers in parallel”. By (P3), an unmeasured uniform superposition is information-free — measuring $Q^{-1/2} \sum_x |x\rangle$ returns a uniformly random x . The algorithm arranges phases so that the wrong answers cancel destructively.

A.2 Unitaries are rotations

Proposition 18 (Unitary = norm-preserving = basis-preserving). *For $U \in \mathcal{L}(\mathcal{H})$ the following are equivalent: (i) $U^\dagger U = \mathbb{1}$; (ii) $\langle Uu|Uv\rangle = \langle u|v\rangle$ for all u, v ; (iii) U maps some orthonormal basis to an orthonormal basis; (iv) U maps every orthonormal basis to an orthonormal basis. In particular $\|U|v\rangle\| = \||v\rangle\|$.*

Proof. (i) $\Rightarrow$ (ii): $\langle Uu|Uv\rangle = \langle u|U^\dagger U|v\rangle = \langle u|v\rangle$. (ii) $\Rightarrow$ (iv) $\Rightarrow$ (iii) are immediate. (iii) $\Rightarrow$ (i): if $\{|e_k\rangle\}$ and $\{U|e_k\rangle\}$ are both orthonormal then $\langle e_j|U^\dagger U|e_k\rangle = \delta_{jk} = \langle e_j|\mathbb{1}|e_k\rangle$ for all j, k , so $U^\dagger U = \mathbb{1}$. $\square$

So a unitary is exactly a rotation of the state sphere: it preserves lengths and angles, and as a real map on $\mathbb{R}^{2d}$ it is orientation-preserving, since $\det_{\mathbb{R}} U = |\det_{\mathbb{C}} U|^2 = 1$. This is why (P2) admits nothing else: among linear maps the unitaries are precisely those carrying a legitimate state description to another, i.e. preserving $\sum_x |\alpha_x|^2 = 1$. (Drop linearity and Wigner’s theorem leaves exactly one further family, the antiunitaries, which no gate implements.) Two consequences:

Corollary 19. *Let U be unitary. (a) U has an orthonormal eigenbasis (spectral theorem for normal operators). (b) Every eigenvalue has modulus 1, hence is $e^{2\pi i\phi}$ for a unique $\phi \in [0, 1)$. (c) If $U|u\rangle = e^{2\pi i\phi}|u\rangle$ then $U^m|u\rangle = e^{2\pi im\phi}|u\rangle$ for all $m \in \mathbb{N}$.*

Proof. (a) is the spectral theorem, applicable since $U^\dagger U = U U^\dagger$. (b) $\||u\rangle\| = \|U|u\rangle\| = |\lambda| \||u\rangle\|$ by Proposition 18. (c) Iterate. $\square$

Part (b) is the starting point: a unitary’s eigenvalue is a phase, a point on the unit circle, and §3 shows a phase can be transferred onto a control qubit and read. Part (c) is the mechanism by which one gate reaches deep into the binary expansion of ϕ : doubling m shifts that expansion one place left (§3).

A.3 Tensor products, separable states, entanglement

If $\{|i\rangle\}$ is a basis of $\mathcal{H}_A$ and $\{|j\rangle\}$ of $\mathcal{H}_B$, then $\{|i\rangle \otimes |j\rangle\}$ is a basis of $\mathcal{H}_A \otimes \mathcal{H}_B$, and $\dim(\mathcal{H}_A \otimes \mathcal{H}_B) = \dim \mathcal{H}_A \cdot \dim \mathcal{H}_B$: the dimensions multiply. The tensor product is bilinear, so it distributes over sums, and that one line of arithmetic is what the circuit model rests on:

$$\left(\sum_i \alpha_i |i\rangle\right) \otimes \left(\sum_j \beta_j |j\rangle\right) = \sum_{i,j} \alpha_i \beta_j |i\rangle \otimes |j\rangle. \quad (15)$$

We write $|i\rangle|j\rangle$ or $|ij\rangle$ for $|i\rangle \otimes |j\rangle$. Operators compose the same way: $(A \otimes B)(|u\rangle \otimes |v\rangle) = A|u\rangle \otimes B|v\rangle$, so a gate acting on one register acts as $A \otimes \mathbb{1}$ on the whole.

Definition 20. A state $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ is *separable* (a product state) if $|\psi\rangle = |\alpha\rangle \otimes |\beta\rangle$ for some $|\alpha\rangle \in \mathcal{H}_A$, $|\beta\rangle \in \mathcal{H}_B$; otherwise it is *entangled*.

Three worked examples, all of which occur below.

(a) *The Hadamard layer is separable.* With $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, expanding by (15),

$$H^{\otimes t}|0\rangle^{\otimes t} = \bigotimes_{s=1}^t \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{2^{t/2}} \sum_{x_1, \dots, x_t \in \{0,1\}} |x_1 \cdots x_t\rangle = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle, \quad Q = 2^t.$$

The middle step is exactly (15) with all 2^t cross terms written out; the last is the big-endian bijection $\{0,1\}^t \rightarrow \{0, \dots, Q-1\}$. So the uniform superposition over an exponentially large index set is a *product* of t independent one-qubit states, prepared by t gates. Each “exponentially large superposition prepared cheaply” below is an instance of this identity.

(b) *A phase ramp is still separable.* For any ϕ ,

$$\bigotimes_{j=1}^t \frac{|0\rangle + e^{2\pi i 2^{t-j}\phi} |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} e^{2\pi i x \phi} |x\rangle,$$

because $\prod_j e^{2\pi i x_j 2^{t-j}\phi} = e^{2\pi i \phi \sum_j x_j 2^{t-j}} = e^{2\pi i x \phi}$: the exponents add exactly as the binary place values do. This is Theorem 8 and, read backwards, Theorem 30; they are one identity read in opposite directions.

(c) *The oracle entangles.* After the modular exponentiation the state is $Q^{-1/2} \sum_x |x\rangle |a^x \bmod N\rangle$. This is *not* separable: if it equalled $|\alpha\rangle \otimes |\beta\rangle$ then the second register would be the same vector $|\beta\rangle$ for every x , whereas $a^0 = 1$ and $a^1 = a$ differ (assuming $r \geq 2$; for $r = 1$ the state is trivially separable and there is nothing to find). Equivalently, measuring the second register changes the distribution of the first. Entanglement here is the correlation that carries the period. On its own, though, it buys nothing: by Gottesman–Knill [22] highly entangled Clifford circuits are classically simulable. The distribution is concentrated by (P5).

A.4 The gates used, defined

Let $\{|0\rangle, |1\rangle\}$ be the computational basis. Single-qubit gates, as matrices in that basis:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i / 2^k} \end{pmatrix}.$$

X is the NOT gate, $X|b\rangle = |b \oplus 1\rangle$. H is the one-qubit Fourier transform: it maps the computational basis to the “phase” basis $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$ and back ($H^2 = \mathbb{1}$); it converts *amplitude* into *phase* and, applied again, phase back into amplitude, and the QFT does the same on t qubits at once.

The R_k are the *phase gates*: $R_k|0\rangle = |0\rangle$ and $R_k|1\rangle = e^{2\pi i / 2^k} |1\rangle$ — they rotate the relative phase of the $|1\rangle$ branch by $2\pi / 2^k$ and do nothing else. The first three have names:

$$R_1 = Z \text{ (phase } \pi, |1\rangle \mapsto -|1\rangle), \quad R_2 = S = \text{diag}(1, i) \text{ (phase } \frac{\pi}{2}), \quad R_3 = T = \text{diag}(1, e^{i\pi/4}) \text{ (phase } \frac{\pi}{4}), \quad (16)$$

so $S = Z^{1/2}$, $T = S^{1/2}$, $T^2 = S$, $S^2 = Z$, $Z^2 = \mathbb{1}$. Note R_k is *diagonal*: it never moves probability between $|0\rangle$ and $|1\rangle$, so measuring immediately after it changes nothing. It affects only the interference pattern produced later, so an H must come before and after. The T gate’s cost is a fault-tolerance matter: $\{H, S, \text{CNOT}\}$ generate the Clifford group, which is classically simulable [22] and cheap to protect, whereas T lies outside it and must be supplied by a non-Clifford resource state — magic-state distillation, or, in current estimates, cultivation (§7). T -count is therefore the currency in which §7’s estimates are quoted, and an *arbitrary*-angle rotation costs $\approx 3 \log_2(1/\epsilon)$ T gates by the optimal *ancilla-free* synthesis of [13] (about $1.15 \log_2(1/\epsilon)$ expected with ancillas and repeat-until-success) — one to two orders of magnitude more than a Toffoli, which is why §C drops small rotations and §E avoids them.

Multi-qubit gates are defined by their action on basis states and extended linearly:

$$\text{CNOT}|c\rangle|b\rangle = |c\rangle|b \oplus c\rangle \quad (\text{controlled-NOT; } c \text{ control, } b \text{ target})$$

$$\begin{aligned}
\text{TOF}|c_1\rangle|c_2\rangle|b\rangle &= |c_1\rangle|c_2\rangle|b \oplus c_1c_2\rangle && (\text{Toffoli, ccNOT}) \\
c\text{-}U|c\rangle|\psi\rangle &= |c\rangle U^c|\psi\rangle && (\text{controlled-}U, c \in \{0,1\}) \\
\text{SWAP}|a\rangle|b\rangle &= |b\rangle|a\rangle, \quad \text{CSWAP}|c\rangle|a\rangle|b\rangle = |c\rangle \text{SWAP}^c(|a\rangle|b\rangle) && (\text{Fredkin})
\end{aligned} \tag{17}$$

Each permutes the computational basis, hence is unitary by Proposition 18(iii). TOF computes the logical AND reversibly ($b = 0$ leaves the target in $|c_1 \wedge c_2\rangle$) and is universal for reversible classical computation — so all the arithmetic of §E is Toffoli circuits, and Toffoli count is the natural cost measure. A Toffoli costs 7 T gates in general, but a Toffoli that is later uncomputed — a *temporary logical-AND* — costs only 4 T for the compute/uncompute *pair*, with the uncomputation done by an X -basis measurement and a classically conditioned CZ, hence Clifford [12]. Almost every Toffoli in §E is of this kind.

“Control” and “target” are naming conventions. For $U = \text{diag}(1, e^{i\theta})$ — diagonal *and* fixing $|0\rangle$ — the two are interchangeable: $c\text{-}Z = \text{diag}(1, 1, 1, -1)$ is symmetric under exchanging its qubits, and so is every $c\text{-}R_k$. Diagonality alone does not suffice ($c\text{-}(-Z) = \text{diag}(1, 1, -1, 1)$ is not symmetric): controlling a gate promotes its global phase to a relative one, which is the mechanism of §3. Even CNOT reverses under a change of basis: $(H \otimes H) \text{CNOT} (H \otimes H)$ is the CNOT with control and target exchanged. A controlled gate is a symmetric interaction; which qubit appears to be acted upon depends on the basis you describe it in. Phase kickback (§3) is an instance of this symmetry.

Appendix B The cyclic shift, its eigenvectors, and circulants

This section is pure linear algebra; no quantum mechanics enters. The claim to establish: the Fourier basis is the unique (up to phases) eigenbasis of the shift, hence the unique basis simultaneously diagonalising every shift-invariant operator — so the spectral decomposition used in this document is necessarily the Fourier one. (That is a statement about this derivation, not about every conceivable procedure for finding a period: Pollard rho and baby-step/giant-step use no Fourier basis at all.) We follow [23, 24].

Lemma 21 (Geometric sum). *For $\omega = e^{2\pi i/M}$ and $j \in \mathbb{Z}$,*

$$\sum_{k=0}^{M-1} \omega^{jk} = M \cdot [j \equiv 0 \pmod{M}], \quad \text{equivalently} \quad \sum_{k=0}^{M-1} \omega^{sk} \overline{\omega^{s'k}} = M \delta_{ss'}. \tag{18}$$

Proof. If $j \equiv 0$ every term is 1. Otherwise $\omega^j \neq 1$ and the geometric series sums to $(\omega^{jM} - 1)/(\omega^j - 1) = 0$. Geometrically: M equally spaced unit steps close a polygon. $\square$

All the cancellations below reduce to (18): orthonormality of the Fourier basis, unitarity of QFT, the fact that $|1\rangle$ is the uniform superposition of eigenvectors (Theorem 11), the exactness of phase estimation on dyadic phases (Theorem 10), and the exact zeros of the output distribution (§3.3).

B.1 The shift, and the operators that commute with it

Let $\omega := \omega_M = e^{2\pi i/M}$ and let S_M be the cyclic shift on $\mathbb{C}^M$, $S_M|k\rangle = |k+1 \pmod{M}\rangle$. A matrix C is *circulant* if $C_{k\ell}$ depends only on $k - \ell \pmod{M}$; equivalently $C = C_b = \sum_m b_m S_M^m$ for a *symbol* $b \in \mathbb{C}^M$ (its first column).

Lemma 22 (Circulant $\Leftrightarrow$ shift-invariant). *$CS_M = S_MC$ if and only if C is circulant.*

Proof. With $(S_M)_{jk} = [j = k+1]$ one computes $(S_MC)_{ij} = C_{i-1,j}$ and $(CS_M)_{ij} = C_{i,j+1}$. So $S_MC = CS_M$ iff $C_{i-1,j} = C_{i,j+1}$ for all i, j , i.e. iff $C_{ij} = C_{i+1,j+1}$: the entries are constant along wrapped diagonals. Setting $b_i := C_{i,0}$ gives $C_{ij} = b_{i-j}$. The converse is immediate from $S_M^m S_M^{m'} = S_M^{m+m'}$. $\square$

Lemma 23 (Simple spectrum forces a common eigenbasis). *If $A \in \mathcal{L}(\mathbb{C}^M)$ has M distinct eigenvalues with eigenvectors $v_1, \dots, v_M$ (so every eigenspace is a line) and $AB = BA$, then every v_i is also an eigenvector of B .*

Proof. $A(Bv_i) = B(Av_i) = \lambda_i(Bv_i)$, so Bv_i lies in the λ_i -eigenspace of A , which is one-dimensional; hence $Bv_i = \gamma_i v_i$. $\square$

Together: *one* basis diagonalises the whole commuting family of shift-invariant operators at once, and it is determined by S_M alone. So we solve for it.

B.2 Why the eigenvalues must be roots of unity

Theorem 24 (Spectrum of the shift). *The eigenvalues of S_M on $\mathbb{C}^M$ are exactly $\omega^0, \omega^1, \dots, \omega^{M-1}$, each simple, with normalised eigenvectors*

$$|u_s\rangle = \frac{1}{\sqrt{M}} \sum_{k=0}^{M-1} \omega^{-sk} |k\rangle, \quad S_M |u_s\rangle = \omega^s |u_s\rangle. \tag{19}$$

Proof. Seek $S_M v = \lambda v$ with $v = \sum_k c_k |k\rangle$. Since $S_M v = \sum_k c_k |k+1\rangle = \sum_k c_{k-1} |k\rangle$, the eigenvalue equation reads $c_{k-1} = \lambda c_k$ for all k , i.e. $c_k = \lambda^{-k} c_0$. Now impose *that the cycle closes*: the index runs in $\mathbb{Z}/M\mathbb{Z}$, so $c_0 = c_M = \lambda^{-M} c_0$, forcing

$$\lambda^M = 1$$

These are M distinct values, so the spectrum is simple. Since S_M permutes an orthonormal basis it is unitary, so $|\lambda| = 1$ automatically (Corollary 19(b)) and normalisation gives $|c_k| = M^{-1/2}$, which is (19). Orthonormality is Lemma 21. $\square$

The eigenvalues are roots of unity because the cycle closes. Nothing analytic entered — no Fourier analysis was assumed. The relation $\lambda^M = 1$ is the operator identity $S_M^M = \mathbb{1}$ read on an eigenvector. Two readings of the same statement, both used below:

Group-theoretic. The $|u_s\rangle$ are the *conjugate* characters of $\mathbb{Z}/M\mathbb{Z}$: writing $\chi_s(k) = \omega^{sk}$, one has $\sqrt{M} u_s(k) = \omega^{-sk} = \overline{\chi_s(k)} = \chi_{-s}(k)$. A character turns *addition of its argument into multiplication of its value*, $\chi_s(k+t) = \chi_s(k)\chi_s(t)$, so translating it can only rescale it: a character is automatically an eigenvector of translation. With $S_M |k\rangle = |k+1\rangle$, which acts on coefficient vectors by $(Sv)_k = v_{k-1}$, shifting by t multiplies χ_s by $\chi_s(-t) = \omega^{-st}$; applied to $|u_s\rangle = \chi_{-s}/\sqrt{M}$ this gives ω^{+st} , matching (19). This explains why the Fourier basis diagonalises translation; the sign is a common source of convention errors.

Ring-theoretic (§B.4). An algebra homomorphism $\varphi : \mathbb{C}[x]/(x^M - 1) \rightarrow \mathbb{C}$ is determined by $z = \varphi(x)$, and $z^M = \varphi(x^M) = \varphi(1) = 1$ forces $z \in \mu_M$, the M -th roots of unity. This is again $\lambda^M = 1$, obtained without computation.

B.3 The DFT as a change of basis; convolution

Definition 25. The (unitary) Fourier matrix is $F := \sum_s |s\rangle\langle u_s|$, i.e. $F_{sk} = \omega^{sk}/\sqrt{M}$; its rows are the conjugated waves $\langle u_s|$; equivalently the columns of $F^\dagger$ are the waves $|u_s\rangle$ themselves. So F is the *analysis* map (signal $\mapsto$ its coordinates in the wave basis) and $F^\dagger$ the *synthesis* map $c \mapsto \sum_s c_s |u_s\rangle$.

Theorem 26 (Diagonalisation and the convolution theorem). *F is unitary, and*

$$FS_M F^\dagger = D := \text{diag}(1, \omega, \dots, \omega^{M-1}), \quad FC_b F^\dagger = \text{diag}(\hat{b}_0, \dots, \hat{b}_{M-1}), \quad \hat{b}_s = \sum_m b_m \omega^{sm}, \quad (20)$$

so the eigenvalue list of a circulant is the DFT of its symbol. Moreover applying a circulant is circular convolution, $(C_a x)_k = \sum_\ell a_{k-\ell} x_\ell =: (a \circledast x)_k$, and

$$\widehat{a \circledast x} = \hat{a} \odot \hat{x} \quad (\odot = \text{entrywise product}). \quad (21)$$

Proof. Unitarity is orthonormality of $\{|u_s\rangle\}$. For (20), apply to $|s\rangle$: $FS_M F^\dagger |s\rangle = FS_M |u_s\rangle = \omega^s F |u_s\rangle = \omega^s |s\rangle$. For a circulant, $C_b |u_s\rangle = \sum_m b_m S^m |u_s\rangle = (\sum_m b_m \omega^{sm}) |u_s\rangle = \hat{b}_s |u_s\rangle$. (Alternatively: C_b commutes with S_M by Lemma 22, S_M has all eigenvalues distinct, so Lemma 23 forces the $|u_s\rangle$ to be its eigenvectors.) The convolution identity is immediate from $(C_a)_{k\ell} = a_{k-\ell}$; then $\widehat{a \circledast x} = \widehat{C_a x}$ and (20) says C_a acts on the s -th Fourier coordinate as multiplication by $\hat{a}_s$, which is (21). $\square$

Read (20) from right to left and it is an *algorithm*: to apply a shift-invariant operator, transform, multiply pointwise, transform back — the $O(M \log M)$ route that replaces an $O(M^2)$ matrix–vector product; its quantum analogue is the Draper adder [6], which appears in Beauregard’s circuit [10].

B.4 Optional: an algebraic interpretation via coefficient and evaluation form

Nothing above or in Part I depends on what follows. The linear algebra is complete: the shift has all eigenvalues distinct, the Fourier matrix diagonalises it, and every circulant follows. This subsection recovers the same facts from ring theory, which explains *why* roots of unity appear and connects the transform to the NTT.

Everything above becomes an accounting identity once one identification is made. Let $\mathcal{R}_M := \mathbb{C}[x]/(x^M - 1)$ and identify $a = (a_0, \dots, a_{M-1}) \in \mathbb{C}^M$ with the polynomial $p_a(x) = \sum_k a_k x^k$. The relation $x^M = 1$ is the wrap-around: $x \cdot p_a = p_{S_M a}$, so under the dictionary

$$S_M = \text{multiplication by } x, \quad C_b = \text{multiplication by } p_b(x), \quad p_a p_b \equiv p_{a \circledast b} \pmod{x^M - 1}. \quad (22)$$

Circular convolution is polynomial multiplication written in coefficient coordinates.

Theorem 27 (DFT = evaluation = CRT). *Let $E : \mathcal{R}_M \rightarrow \mathbb{C}^M$, $E(p) = (p(\omega^0), p(\omega^1), \dots, p(\omega^{M-1}))$. Then $E(p_a)_s = \sum_k a_k \omega^{sk} = \hat{a}_s$: the matrix of E is the (unnormalised) DFT, the Vandermonde matrix at the nodes μ_M . Moreover the factorisation $x^M - 1 = \prod_s (x - \omega^s)$ into distinct linear factors induces, by the Chinese Remainder Theorem, a ring isomorphism*

$$\mathbb{C}^M \cong \mathbb{C}[x]/(x^M - 1) \xrightarrow{\sim} \prod_{s=0}^{M-1} \mathbb{C}[x]/(x - \omega^s) \cong \mathbb{C}^M, \quad p \mapsto (p(\omega^s))_s, \quad (23)$$

whose inverse is Lagrange interpolation at μ_M — which, because the nodes form a group, collapses to the conjugate transform $a_k = \frac{1}{M} \sum_s \hat{a}_s \omega^{-sk}$.

Proof. The linear factors $x - \omega^s$ are pairwise coprime, so CRT gives (23) with s -th component reduction mod $(x - \omega^s)$, i.e. $p \mapsto p(\omega^s)$; this descends to $\mathcal{R}_M$ precisely because the modulus vanishes at the nodes, $(\omega^s)^M - 1 = 0$. Bijectivity is concrete: $\sum_k \omega^{sk} \omega^{-s'k} = M \delta_{ss'}$ by Lemma 21, which is both the inversion formula and the unitarity of F . $\square$

Corollary 28 (Convolution theorem). $\widehat{a \circledast b}_s = p_{a \circledast b}(\omega^s) = (p_a p_b)(\omega^s) = p_a(\omega^s) p_b(\omega^s) = \hat{a}_s \hat{b}_s$.

Proof. The dictionary (22) gives $p_{a \circledast b} \equiv p_a p_b$, and evaluation at a fixed point is multiplicative. The CRT isomorphism transports the awkward product $\circledast$ to the obvious one $\odot$. $\square$

Why $(xp)(\omega^s) = \omega^s p(\omega^s)$ is the statement $FS_M F^\dagger = D$

This step reduces the spectral theorem for the shift to bookkeeping, so we set it out in four observations, none of which requires a computation.

(1) *What the operator is.* By the dictionary (22), the shift S_M is the operator “multiply by x ” on $\mathcal{R}_M = \mathbb{C}[x]/(x^M - 1)$. Nothing about waves or transforms has been assumed; this is just the statement that $x \cdot p_a = p_{S_M a}$, i.e. that multiplying by x pushes every coefficient up one degree and recycles the top one because $x^M = 1$.

(2) *What the coordinates are.* The evaluation map $E(p) = (p(\omega^0), \dots, p(\omega^{M-1}))$ of Theorem 27 is a *change of coordinates* on $\mathcal{R}_M$: it replaces the coefficient list a by the value list $\hat{a}$. The s -th coordinate of a ring element is, by definition, “its value at ω^s ”.

(3) *The computation, which uses only that evaluation is a ring homomorphism.* Ask what multiplication by x does to the s -th coordinate. Evaluation is multiplicative, and the polynomial x evaluates at ω^s to the number ω^s ; hence for every p ,

$$E(x \cdot p)_s = (xp)(\omega^s) = \underbrace{x|_{x=\omega^s}}_{=\omega^s} \cdot p(\omega^s) = \omega^s E(p)_s. \quad (24)$$

Read the two ends: the s -th coordinate of the output is ω^s times the s -th coordinate of the input, *for every p and with no mixing between coordinates*. An operator that scales each coordinate by a fixed number and mixes nothing is, by definition, *diagonal* — with diagonal entries $\omega^0, \dots, \omega^{M-1}$. So in evaluation coordinates the shift is $D = \text{diag}(\omega^s)$, and it is diagonal by the definition of evaluation: knowing p only through its value at ω^s , all multiplication by x can do to that datum is multiply it by ω^s .

(4) *Rewriting (3) as a matrix identity.* “ T acts as D in the coordinates supplied by E ” means $E T E^{-1} = D$, since a change of basis conjugates operators. Equation (24) says exactly $ES_M = DE$, i.e.

$$E S_M E^{-1} = D. \quad (25)$$

Finally $E = \sqrt{M} F$ as matrices (Definition 25 carries the $M^{-1/2}$ that Theorem 27 omits), and a scalar cancels against its inverse under conjugation, so $FS_M F^{-1} = ES_M E^{-1} = D$; and $F^{-1} = F^\dagger$ because F is unitary. Hence

$$FS_M F^\dagger = D.$$

$$\begin{array}{ccc} a \in (\mathbb{C}^M, \circledast) & \xrightarrow{S_M = \text{multiply by } x} & S_M a \\ E \downarrow \text{evaluate at } \mu_M & & \downarrow E \\ \hat{a} \in (\mathbb{C}^M, \odot) & \xrightarrow{D = \text{diag}(\omega^s)} & \hat{a} \odot (\omega^s)_s \end{array}$$

The square commutes, which is the identity $FS_M F^\dagger = D$: going right-then-down equals going down-then-right. Read right-to-left, $S_M = F^\dagger D F$ becomes the instruction “analyse, twiddle each coordinate, resynthesise”, which is how the identity is used computationally.

Three consequences. (i) *Eigenvalues.* A diagonal matrix wears its spectrum on its sleeve, so the eigenvalues of S_M are $\{\omega^s\}$ — recovering Theorem 24 with no eigenvalue equation solved. The condition $\lambda^M = 1$ of that theorem is here the single fact $(\omega^s)^M = 1$, i.e. that ω^s is a *root* of the modulus $x^M - 1$: the eigenvalues are the roots of unity because the eigenvalues are exactly the points at which we evaluate, and those points are the roots of the polynomial we quotiented by.

(ii) *Eigenvectors.* The s -th evaluation coordinate is extracted by the Lagrange idempotent $\ell_s(x) = \frac{1}{M} \sum_k \omega^{-sk} x^k$ (the element with $\ell_s(\omega^t) = \delta_{st}$), whose coefficient vector is $M^{-1/2}|u_s\rangle$. So the Fourier waves (19) are the Lagrange basis in disguise, and “ S_M is diagonal in evaluation coordinates” says that each $|u_s\rangle$ is an eigenvector.

(iii) *All circulants at once.* Nothing in (24) used that the multiplier was x . Multiplication by any p_b acts on the s -th coordinate as the scalar $p_b(\omega^s) = \hat{b}_s$, so $FC_b F^\dagger = \text{diag}(\hat{b})$ — Theorem 26 again, and with it the convolution theorem, since a diagonal operator is exactly a pointwise product. Taking $b = \delta_1$ recovers S_M .

So the spectral theorem for the shift adds nothing beyond the dictionary. Its content is Theorem 27: coefficient form and evaluation form describe the same object; once one sits in evaluation coordinates, S is diagonal *by definition of evaluation*. Theorem 24 reached the same endpoint by solving $\lambda^M = 1$ by hand. The first route shows why the answer could not have come out otherwise; the second shows why it is immediate.

The DFT is a bijection between two ways of writing the same object.

Coefficient form a — the “position” or time-domain representation; the computational basis of a quantum register. *Evaluation form* $\hat{a} = (p_a(\omega^s))_s$ — the “frequency” representation; the Fourier basis.

The transform is the map between them, and it is an *isomorphism of rings*: it carries the awkward product $\otimes$ to the obvious product $\odot$. The convolution theorem, the simultaneous diagonalisation of all circulants, Lagrange interpolation and Parseval are all restatements of this. Its computational payoff is the FFT: the coefficient-to-evaluation bijection, naively an $M \times M$ matrix–vector product costing $O(M^2)$, computable in $O(M \log M)$: when M is smooth — and $M = 2^t$ throughout this document — the CRT can be iterated over a tower of subgroups of μ_M (Cooley–Tukey); for prime M the same bound is reached differently (Rader, Bluestein). The quantum Fourier transform of §C is literally the matrix F of Definition 25 — the evaluation map E normalised by $M^{-1/2}$ — applied to amplitudes instead of coefficients, at a cost of $\Theta(\log^2 M)$ gates against the FFT’s $O(M \log M)$ operations. It applies the Fourier matrix using $\text{poly}(\log M)$ gates, but it does not output a classical list of the M Fourier coefficients — a distinction worth stating, since the opposite reading is a standard misconception.

Nothing in the *algebraic* half of this section used $\mathbb{C}$: the evaluation/CRT picture needs only a commutative ring with a principal M -th root of unity and M invertible. Over $\mathbb{Z}_q$ it is the number-theoretic transform. The lattice standards use its *negacyclic* variant modulo $x^n + 1$, which needs a primitive $2n$ -th root of unity: Dilithium/ML-DSA has one ($q = 8380417 \equiv 1 \pmod{512}$), whereas Kyber/ML-KEM’s $q = 3329$ is $1 \pmod{256}$ but not $\pmod{512}$, so its transform deliberately stops at 128 quadratic factors [25, 26]. The same algebra appears in the lattice-based replacements for RSA.

Appendix C The quantum Fourier transform as a circuit

Definition 29. For $Q = 2^t$ and $\omega_Q = e^{2\pi i/Q}$,

$$\text{QFT}_Q |x\rangle = \frac{1}{\sqrt{Q}} \sum_{y=0}^{Q-1} \omega_Q^{xy} |y\rangle, \quad \text{hence} \quad \text{QFT}_Q^\dagger |x\rangle = \frac{1}{\sqrt{Q}} \sum_{y=0}^{Q-1} \omega_Q^{-xy} |y\rangle.$$

Unitarity is Lemma 21: $\langle x' | \text{QFT}_Q^\dagger \text{QFT}_Q |x\rangle = \frac{1}{Q} \sum_y \omega_Q^{(x-x')y} = \delta_{xx'}$. Note $\text{QFT}_Q |0\rangle = H^{\otimes t} |0\rangle^{\otimes t}$ and $\text{QFT}_2 = H$.

Theorem 30 (Product form). For $Q = 2^t$ and $|j\rangle = |j_1 j_2 \cdots j_t\rangle$ big-endian,

$$\text{QFT}_{2^t} |j_1 \cdots j_t\rangle = \frac{1}{2^{t/2}} \bigotimes_{\ell=1}^t \left(|0\rangle + e^{2\pi i (0.j_{t-\ell+1} \cdots j_t)} |1\rangle \right), \quad 0.j_m \cdots j_t := \sum_{s \geq 1} j_{m+s-1} 2^{-s}. \quad (26)$$

Proof. Write $k = \sum_{\ell=1}^t k_\ell 2^{t-\ell}$, so $k/2^t = \sum_{\ell} k_\ell 2^{-\ell}$. Then

$$\text{QFT} |j\rangle = \frac{1}{2^{t/2}} \sum_k e^{2\pi i j k / 2^t} |k\rangle = \frac{1}{2^{t/2}} \sum_{k_1, \dots, k_t \in \{0,1\}} \prod_{\ell=1}^t e^{2\pi i j k_\ell 2^{-\ell}} |k_1 \cdots k_t\rangle,$$

by multiplicativity of the exponential. The sum over t independent bits of a product factorises, giving $\bigotimes_{\ell} (|0\rangle + e^{2\pi i j 2^{-\ell}} |1\rangle) / 2^{t/2}$. Finally $j 2^{-\ell} \equiv 0.j_{t-\ell+1} \cdots j_t \pmod{1}$, because the contributions of $j_1, \dots, j_{t-\ell}$ to $j 2^{-\ell}$ are integers and drop out of $e^{2\pi i(\cdot)}$. $\square$

Two readings of the same identity. (26) says the QFT of a *basis* state is a *product* state, and so is never entangled. The circuit can therefore be small: the t wires are prepared independently. Read right-to-left, (26) is the kickback ladder (10): a tensor product of $|0\rangle + e^{2\pi i 2^{t-j}\phi} |1\rangle$ is the ramp $\sum_x e^{2\pi i x \phi} |x\rangle$. So the ladder builds $\text{QFT}|Q\phi\rangle$, and $\text{QFT}^\dagger$ is the natural thing to apply to it. Phase estimation is that combination.

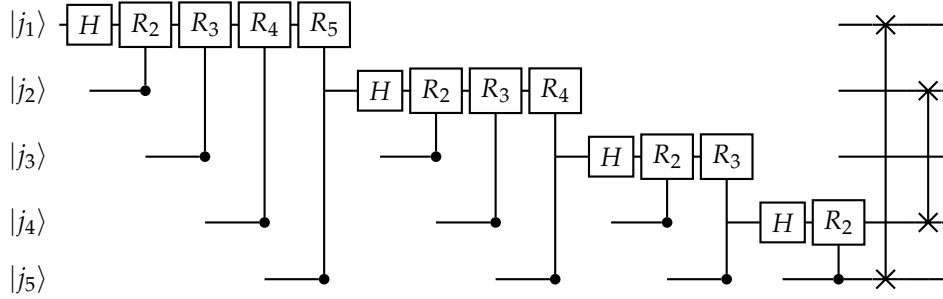
C.1 Building the circuit from (26)

Let $R_k := \text{diag}(1, e^{2\pi i/2^k})$, so $R_1 = Z$, $R_2 = S$, $R_3 = T$.

Lemma 31. Fix s and suppose qubits $s, s+1, \dots, t$ hold the basis state $|j_s j_{s+1} \dots j_t\rangle$. Apply to qubit s : first H , then for $k = 2, \dots, t-s+1$ a $c\text{-}R_k$ with control qubit $s+k-1$ and target qubit s . Then qubit s ends in $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i(0.j_s j_{s+1} \dots j_t)} |1\rangle)$ and qubits $s+1, \dots, t$ are unchanged.

Proof. $H|j_s\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{j_s}|1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.j_s}|1\rangle)$ since $0.j_s = j_s/2$. The control $s+k-1$ is a basis state, so there is no back-action, and $c\text{-}R_k$ multiplies the $|1\rangle$ component by $e^{2\pi i j_{s+k-1}/2^k}$. Accumulating, $\exp(2\pi i \sum_{k=2}^{t-s+1} j_{s+k-1} 2^{-k}) = e^{2\pi i(0.0j_{s+1} \dots j_t)}$, and $0.j_s + 0.0j_{s+1} \dots j_t = 0.j_s j_{s+1} \dots j_t$. $\square$

Order matters. Lemma 31 needs qubits $s+1, \dots, t$ to still be basis states when they serve as controls. So process qubit 1 first, then 2, etc.; once qubit s has been Hadamarded it is no longer a basis state and may not be used as a control.

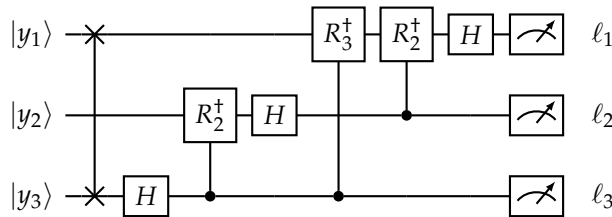


QFT_{2^5} . Each H writes the leading bit of a binary fraction; each $c\text{-}R_k$ appends the k -th bit without disturbing the wire that supplies it. Output factor ℓ of (26) depends on the *tail* $j_{t-\ell+1} \dots j_t$, so the natural circuit emits the qubits reversed; the closing $\lfloor t/2 \rfloor$ swaps repair this (or one relabels classically, which is free).

Theorem 32 (Correctness and cost). *The circuit implements QFT_{2^t} with t Hadamards, $\binom{t}{2}$ controlled rotations and $\lfloor t/2 \rfloor$ swaps: $\frac{t(t+1)}{2} + \lfloor t/2 \rfloor = \Theta(t^2)$ gates, at depth $\Theta(t)$ if commuting rotations are parallelised.*

Proof. Apply Lemma 31 for $s = 1, \dots, t$ in order; its hypothesis is maintained because gates acting on qubit s never alter qubits $s+1, \dots, t$ (controls are unchanged by a controlled gate). Wire s receives one H and $t-s$ rotations, and $\sum_s (t-s) = \binom{t}{2}$. $\square$

The inverse QFT. For any circuit $V = G_L \dots G_1$ one has $V^\dagger = G_1^\dagger \dots G_L^\dagger$: run the diagram backwards and dagger every gate. Here $H^\dagger = H$, $\text{SWAP}^\dagger = \text{SWAP}$, and $R_k^\dagger = \text{diag}(1, e^{-2\pi i/2^k})$. So $\text{QFT}_{2^t}^\dagger$ is: the swap layer; then for $s = t, t-1, \dots, 1$, first the rotations $c\text{-}R_k^\dagger$ ($k = t-s+1, \dots, 2$) and then H on qubit s . Two checks before trusting an implementation: (a) build the $2^t \times 2^t$ matrix for $t = 3, 4$ and compare entrywise with $Q^{-1/2} \omega_Q^{xy}$; (b) verify $\text{QFT}^\dagger \text{QFT} = \mathbb{1}$ numerically. A failure is almost always the swap layer or the bit order.



$\text{QFT}_{2^3}^\dagger$, fully expanded, with the measurement that follows it in every use below. Read the forward circuit backwards: swaps first, then wire 3 (which the forward circuit touched last), then wire 2, then wire 1; on each wire the rotations precede its H , and every rotation angle is negated. The last gate on wire 1 is an H , so the top wire is measured last in the natural ordering — and each qubit's rotations depend only on wires measured before it, the structure that the semiclassical variant of §E.5 exploits to replace all of this by one recycled qubit and classical feed-forward [4].

Theorem 33 (Approximate QFT [5]). *Let AQFT_m omit every $c\text{-}R_k$ with $k > m$. Then $\|\text{QFT}_{2^t} - \text{AQFT}_m\| \leq 2\pi t/2^m$, and AQFT_m uses $O(tm)$ gates. Taking $m = \lceil \log_2(2\pi t/\epsilon) \rceil$ gives error $\leq \epsilon$ with $O(t \log(t/\epsilon))$ gates.*

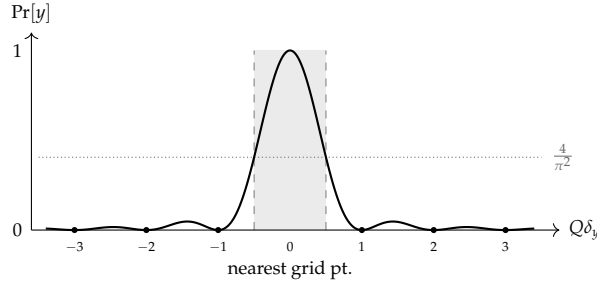


Figure 2: The Fejér kernel (27) as a function of the detuning $Q\delta_y$. Dots mark the exact zeros at non-zero integer detunings; the dashed lines bound the nearest grid point, $|\delta| \leq 1/(2Q)$, where the $4/\pi^2$ guarantee applies.

Proof. $\|R_k - \mathbb{1}\| = |e^{2\pi i/2^k} - 1| \leq 2\pi/2^k$. For products of unitaries $\|V - \tilde{V}\| \leq \sum_\ell \|V_\ell - \tilde{V}_\ell\|$ (telescope $V - \tilde{V} = \sum_\ell V_L \cdots V_{\ell+1}(V_\ell - \tilde{V}_\ell)\tilde{V}_{\ell-1} \cdots \tilde{V}_1$ and use that unitaries have operator norm 1). There are $< t$ omitted rotations per wire and t wires; the geometric tail $\sum_{k>m} 2\pi 2^{-k} = 2\pi 2^{-m}$. $\square$

In the fault-tolerant cost model an arbitrary-angle rotation costs $\approx 3 \log_2(1/\epsilon)$ T gates [13], one to two orders of magnitude more than a Toffoli. Every serious resource estimate uses the AQFT.

Appendix D The exact output distribution, and how many qubits it costs

§3.2 stated that phase estimation concentrates on the grid point nearest $Q\phi$ and quantified the concentration by $4/\pi^2$. Here is the exact distribution, the proof of that bound, the qubit count needed for a prescribed accuracy, and the statement that legitimises feeding in $|1\rangle$ rather than an eigenvector.

Theorem 34 (Exact output distribution: a Fejér kernel). *Put $\delta_\ell := \phi - \ell/Q \bmod 1$, taken in $[-\frac{1}{2}, \frac{1}{2})$. Then*

$$\alpha_\ell = \frac{1}{Q} \cdot \frac{1 - e^{2\pi i Q \delta_\ell}}{1 - e^{2\pi i \delta_\ell}} \quad (\delta_\ell \neq 0), \quad \Pr[\ell] = |\alpha_\ell|^2 = \frac{1}{Q^2} \cdot \frac{\sin^2(\pi Q \delta_\ell)}{\sin^2(\pi \delta_\ell)}, \quad (27)$$

with value 1 at $\delta_\ell = 0$. This is the intensity pattern of a Q -slit diffraction grating.

Proof. The geometric sum in (11) with ratio $e^{2\pi i \delta_\ell} \neq 1$ gives the first display; then use $|1 - e^{i\theta}| = 2|\sin(\theta/2)|$ twice and square. $\square$

Theorem 35 (The $4/\pi^2$ guarantee). *Let b be an integer nearest to $Q\phi$ modulo Q , so $|\delta_b| \leq \frac{1}{2Q}$. Then*

$$\Pr[b] \geq \frac{4}{\pi^2} = 0.4052 \dots \quad (28)$$

Moreover $\Pr[\ell]$ depends on ℓ only through $|\delta_\ell|$ and is non-increasing in it, so b maximises it — uniquely unless $|\delta_b| = \frac{1}{2Q}$ exactly, which is a tie.

Proof. If $\delta_b = 0$ then $\Pr[b] = 1$. Else put $\theta := \pi Q |\delta_b| \in (0, \pi/2]$. From (27), $\sqrt{\Pr[b]} = \frac{1}{Q} \frac{\sin \theta}{|\sin(\pi \delta_b)|}$. Bound the two factors separately: $|\sin x| \leq |x|$ gives $|\sin(\pi \delta_b)| \leq \pi |\delta_b|$; and Jordan's inequality $\sin \theta \geq \frac{2}{\pi} \theta$ on $(0, \pi/2]$ gives $\sin \theta \geq 2Q |\delta_b|$. Hence $\sqrt{\Pr[b]} \geq \frac{1}{Q} \cdot \frac{2Q |\delta_b|}{\pi |\delta_b|} = \frac{2}{\pi}$. For the maximisation: since $Q\delta_\ell = Q\phi - \ell$ differs from $Q\phi$ by an integer, the numerator $\sin^2(\pi Q \delta_\ell) = \sin^2(\pi Q \phi)$ is the same for every ℓ , so $\Pr[\ell] = \sin^2(\pi Q \phi) / (Q^2 \sin^2(\pi \delta_\ell))$ is a function of $|\delta_\ell|$ alone, strictly decreasing in it unless $\sin(\pi Q \phi) = 0$ — the exact case of Theorem 10, where the distribution is a point mass. $\square$

What repetition does and does not buy. Since $4/\pi^2 < \frac{1}{2}$, the guaranteed probability of the nearest grid point is below one half, so ordinary majority (or mode) amplification of that event does not apply, and the mode can in any case be tied. Repetition is still useful, and the algorithm relies on it: independent runs raise the probability that *at least one* usable sample appears, which is exactly what the lcm strategy of §5 exploits. What repetition does not do is sharpen a single estimate; accuracy is bought instead with extra *counting qubits* (Theorem 36), which is exponentially cheap.

Sharpness. At $|\delta_b| = \frac{1}{2Q}$ exactly, $\Pr[b] = (Q \sin \frac{\pi}{2Q})^{-2}$, which decreases to $4/\pi^2$: numerically 0.410533 at $t = 3$, 0.405366 at $t = 6$, 0.405285 at $t = 10$. No constant $> 4/\pi^2$ works uniformly. A parallel computation gives $\Pr[b_0] + \Pr[b_1] \geq 8/\pi^2 = 0.8105 \dots$ for the two grid points straddling $Q\phi$.

Theorem 36 (Precision–qubit trade-off). *Fix $v \geq 1$, $\epsilon \in (0, \frac{1}{2})$ and run QPE with*

$$t = v + \left\lceil \log_2 \left(2 + \frac{1}{2\epsilon} \right) \right\rceil \quad (29)$$

counting qubits. Then $|\ell/Q - \phi| < 2^{-v}$ (as a distance mod 1) with probability $\geq 1 - \epsilon$.

Proof. Let $b = \lfloor Q\phi \rfloor$ and $\delta = \phi - b/Q \in [0, 2^{-t})$. Index outcomes by $\ell' = (b + \ell) \bmod Q$ with $-Q/2 < \ell \leq Q/2$; then $\alpha_\ell = \frac{1}{Q} \frac{1 - e^{2\pi i(Q\delta - \ell)}}{1 - e^{2\pi i(\delta - \ell/Q)}}$. The numerator has modulus ≤ 2 ; and $|1 - e^{i\theta}| \geq 2|\theta|/\pi$ for $|\theta| \leq \pi$ applied with $\theta = 2\pi(\delta - \ell/Q)$ gives $|1 - e^{2\pi i(\delta - \ell/Q)}| \geq 4|\delta - \ell/Q|$, so $|\alpha_\ell| \leq \frac{1}{2|Q\delta - \ell|}$. For an integer $e \geq 2$, using $0 \leq Q\delta < 1$,

$$\Pr[|\ell| > e] \leq \frac{1}{4} \left[\sum_{\ell \leq -(e+1)} \frac{1}{(Q\delta - \ell)^2} + \sum_{\ell \geq e+1} \frac{1}{(Q\delta - \ell)^2} \right] \leq \frac{1}{2} \sum_{j \geq e} \frac{1}{j^2} \leq \frac{1}{2} \int_{e-1}^{\infty} \frac{dx}{x^2} = \frac{1}{2(e-1)}.$$

Take $e := 2^{t-\nu} - 1$. If $|\ell| \leq e$ then $|\ell'/Q - \phi| \leq |\ell|/Q + \delta < (2^{t-\nu} - 1)/2^t + 2^{-t} = 2^{-\nu}$. It suffices that $\frac{1}{2(e-1)} \leq \varepsilon$, i.e. $2^{t-\nu} \geq 2 + \frac{1}{2\varepsilon}$, which is (29). $\square$

(29) splits the cost cleanly: ν qubits buy *resolution* $2^{-\nu}$, and $\lceil \log_2(2 + \frac{1}{2\varepsilon}) \rceil$ extra qubits buy *confidence* $1 - \varepsilon$ (six qubits for $\varepsilon = 10^{-2}$, nineteen for 10^{-6}). Confidence is exponentially cheap; resolution costs more, since each extra bit doubles the applications of U .

Theorem 37 (Input that is not an eigenvector). *Let $U = \sum_u e^{2\pi i \phi_u} |u\rangle\langle u|$ and run QPE on $|0\rangle^{\otimes t} \otimes |\psi\rangle$ with $|\psi\rangle = \sum_u c_u |u\rangle$. Then $|\Psi_3\rangle = \sum_u c_u (\sum_\ell \alpha_\ell^{(u)} |\ell\rangle) \otimes |u\rangle$ and*

$$\Pr[\ell] = \sum_u |c_u|^2 |\alpha_\ell^{(u)}|^2. \quad (30)$$

That is: the algorithm behaves exactly as if $|u\rangle$ had first been drawn with probability $|c_u|^2$ and QPE run on it.

Proof. The circuit is linear and Theorem 9 computed its action on each $|0\rangle^{\otimes t} |u\rangle$; superpose. For the probability, the $|u\rangle$ are orthonormal, so cross terms vanish. Equivalently: a hypothetical measurement of the second register in $\{|u\rangle\}$ commutes with every gate of the circuit (each is supported on the counting register, or is a c -UP, diagonal in $\{|u\rangle\}$), so it may be performed after the circuit, where it cannot influence the already-recorded outcome. $\square$

Appendix E Reversible modular arithmetic and the controlled- U tower

Three ingredients: the Hadamard layer (t gates), the inverse QFT ($\mathbb{S}C$, $\Theta(t^2)$ or $O(t \log t)$ gates), and the controlled modular multiplications, which dominate the gate count (in the RSA-2048 estimate of §7, by a factor $\approx 3 \times 10^4$). This appendix builds the last, bottom-up.

E.1 Level 0–1: reversible addition

Schoolbook binary addition [9] is $s_i = a_i \oplus b_i \oplus c_i$, $c_{i+1} = \text{MAJ}(a_i, b_i, c_i) = a_i b_i \oplus a_i c_i \oplus b_i c_i$, $c_0 = 0$. The sum bits are $\mathbb{F}_2$ -linear given the carries, hence free (CNOTs); the carries are quadratic, hence cost one Toffoli-like gate each; and they form a chain, hence depth $\Theta(n)$. Define, on an ordered triple of wires (c, b, a) ,

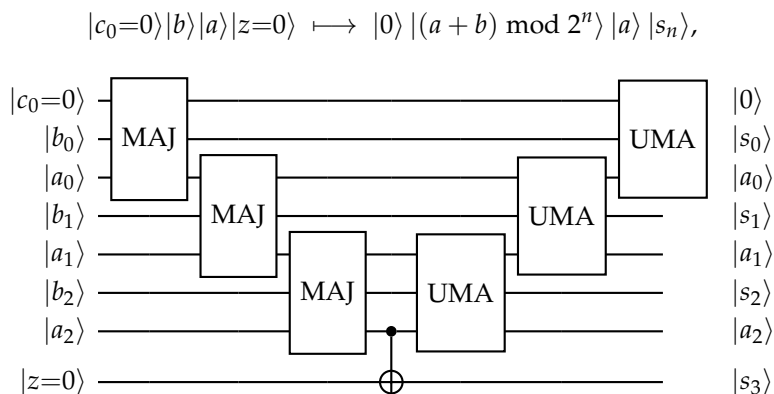
$$\text{MAJ} : \text{CNOT}(a \rightarrow b), \text{CNOT}(a \rightarrow c), \text{TOF}(c, b \rightarrow a); \quad \text{UMA} : \text{TOF}(c, b \rightarrow a), \text{CNOT}(a \rightarrow c), \text{CNOT}(c \rightarrow b).$$

Lemma 38. *MAJ and UMA are partial inverses of one another:*

$$\text{MAJ} : (c, b, a) \mapsto (c \oplus a, b \oplus a, \text{MAJ}(a, b, c)), \quad \text{UMA} : (c \oplus a, b \oplus a, \text{MAJ}(a, b, c)) \mapsto (c, a \oplus b \oplus c, a).$$

Proof. Track the three assignments. For MAJ the only non-trivial step is $a \mapsto a \oplus (b \oplus a)(c \oplus a) = a \oplus bc \oplus ab \oplus ac \oplus a = ab \oplus ac \oplus bc$, using $a \cdot a = a$ and $a \oplus a = 0$ over $\mathbb{F}_2$. For UMA, the Toffoli gives $\text{MAJ}(a, b, c) \oplus (b \oplus a)(c \oplus a) = a$ by the same computation read backwards; then $C \mapsto C \oplus a = c$ and $B \mapsto B \oplus c = a \oplus b \oplus c$. $\square$

Chaining n MAJ blocks propagates all carries; one CNOT copies out the top carry; then n UMA blocks in *reverse* order simultaneously uncompute the carries and write the sum bits. This is the CDKM ripple-carry adder:



CDKM adder, $n = 3$ [9]. Wires interleave $c_0, b_0, a_0, b_1, a_1, \dots, b_{n-1}, a_{n-1}, z$ so every block touches three adjacent wires. The descending MAJ chain computes all carries; the single CNOT copies out the top carry; the ascending UMA chain simultaneously *uncomputes* the carries and writes the sum bits — compute-copy-uncompute with the uncomputation doing useful work on the way back, which is why the circuit has zero garbage.

This costs, in place with operand a preserved and **zero garbage**, $2n$ Toffoli, $4n+1$ CNOT, $2n+2$ qubits ($4n$ T gates using temporary-AND, since each carry is computed and later uncomputed). Running it backwards subtracts; a comparator $|a\rangle|b\rangle|0\rangle \mapsto |a\rangle|b\rangle|[a \geq b]\rangle$ is the MAJ chain run on a and $\bar{b}$ with carry-in 1 — its top carry is the carry-out of $a - b + 2^n$, hence 1 iff $a \geq b$ — then one CNOT to copy it out, then the chain reversed: n compute/uncompute Toffoli pairs, $4n$ T . Every addition inside Shor’s algorithm has a *compile-time classical* operand, which deletes or fixes roughly half the gates before emission.

E.2 Level 2: modular addition by a classical constant

Work with $n+1$ -qubit registers so intermediate values up to $2N$ fit.

Theorem 39 (Beauregard’s modular adder [10], after Vedral–Barenco–Ekert; here instantiated with the ripple-carry adder of §E; Beauregard’s original uses a Fourier-basis adder, where the sign bit must be extracted by a QFT[†]/QFT pair around step (3)). For $0 \leq a < N$ the following seven blocks implement $|x\rangle \mapsto |(x+a) \bmod N\rangle$ for all $0 \leq x < N$, using five additions of classical constants and one ancilla t returned to $|0\rangle$:

(1) ADD(a)	$x + a < 2N$	(5) SUB(a)	$\mapsto R - a$
(2) SUB(N)	$\mapsto x + a - N$, possibly negative	(6) X on t ; CNOT sign $\rightarrow t$	$t \mapsto 0$
(3) CNOT sign qubit $\rightarrow t$	$t = [x + a < N]$	(7) ADD(a)	$\mapsto R$
(4) ADD(N) controlled on t	$\mapsto R := (x + a) \bmod N$		

Proof. Steps 1–2 give $x + a - N$, negative exactly when $x + a < N$; in two’s complement the sign qubit is $[x + a < N]$, so step 3 records it and step 4 restores $R \in [0, N)$. The remaining step is clearing the ancilla. Since $x < N$,

$$t = 1 \iff x + a < N \iff R = x + a \geq a, \quad t = 0 \iff R = x + a - N < a,$$

so $t = [R \geq a]$ is recomputable from the register alone. Step 5 forms $R - a$, whose sign qubit is $[R < a] = \neg t$; the X makes $t \mapsto \neg t$ and the CNOT then gives $\neg t \oplus \neg t = 0$. Step 7 restores R . $\square$

This pattern — *every block ends by recomputing its own flag from its own output* — keeps the whole tower garbage-free, which by (P5) is required for correctness.

E.3 Level 3–4: multiplication, and the controlled multiplier

Theorem 40 ([10]). Let $\gcd(a, N) = 1$ and $\bar{a} := a^{-1} \bmod N$ (classical, extended Euclid). Then

$$\text{CMULT}(a) : |c\rangle|x\rangle|b\rangle \mapsto |c\rangle|x\rangle|b + c \cdot ax \bmod N\rangle \quad \text{is} \quad \prod_{i=0}^{n-1} \text{ADD}_{c, x_i}^{\bmod N}(2^i a \bmod N), \quad (31)$$

n doubly-controlled modular additions of compile-time constants; and

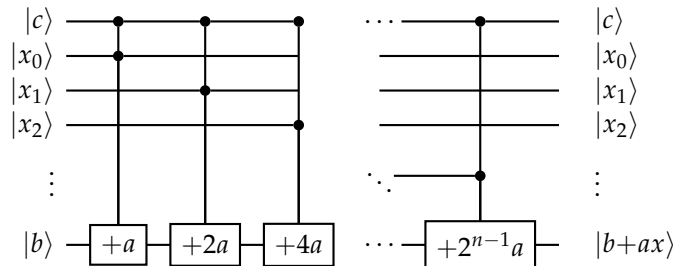
$$c\text{-}U_a = \text{CMULT}(\bar{a})^\dagger \cdot c\text{-SWAP} \cdot \text{CMULT}(a) \quad (32)$$

implements $|c\rangle|x\rangle|0\rangle \mapsto |c\rangle|a^c x \bmod N\rangle|0\rangle$ for all $0 \leq x < N$.

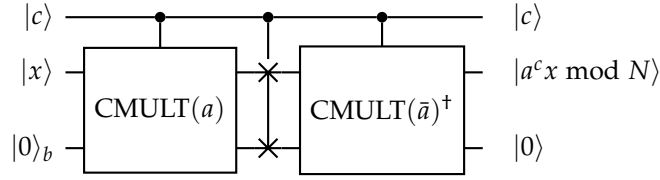
Proof. For (31): $ax \bmod N = (\sum_i x_i (2^i a \bmod N)) \bmod N$, and the i -th adder fires exactly when $x_i = 1$; each adder reduces, so all partial sums stay $< N$. For (32), take $c = 1$ and follow the register pair:

$$|x\rangle|0\rangle \xrightarrow{\text{CMULT}(a)} |x\rangle|ax\rangle \xrightarrow{\text{SWAP}} |ax\rangle|x\rangle \xrightarrow{\text{CMULT}(\bar{a})^\dagger} |ax\rangle|x - \bar{a}(ax)\rangle = |ax\rangle|0\rangle,$$

since $\bar{a}a \equiv 1$. For $c = 0$ every factor is the identity. Both branches leave the scratch register in $|0\rangle$, so the output is unentangled and $c\text{-}U_a$ is realised exactly. $\square$



CMULT(a) expanded (31). Each box is one modular addition of the compile-time constant $2^i a \bmod N$, doubly controlled by the outer control c and the bit x_i ; every partial sum is reduced mod N , so the invariant $b < N$ needed by Theorem 39 holds throughout. The bit x_i selects whether its multiple is added: the construction is binary long multiplication written reversibly.



The circuit contains no controlled QFT and no control on the constant $\bar{a}$; the inverse is classical. The controlled SWAP is n Fredkin gates sharing the single control. An *in-place* multiplier exists only because a is invertible mod N ; this is checked classically beforehand, and if it fails one has already factored N .

E.4 Where the order of magnitude is won: do not control every gate

Lemma 41 (Control propagation [10]). *Let $V = G_L \cdots G_1$ and $S \subseteq \{1, \dots, L\}$; let V_S replace G_j by $c\text{-}G_j$ for $j \in S$. Then $V_S = c\text{-}V$ on a subspace W of admissible inputs iff the residue circuit $R_S := \prod_{j \notin S} G_j$ equals $\mathbb{1}$ on W .*

Proof. $V_S = |0\rangle\langle 0| \otimes R_S + |1\rangle\langle 1| \otimes V$, while $c\text{-}V = |0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes V$. $\square$

Two patterns exhaust the applications. **(a) Conjugation:** if $V = ABA^\dagger$ then $S = \{B\}$ suffices, since $R_S = AA^\dagger = \mathbb{1}$. A *conjugating layer never needs the control* — e.g. the $\text{QFT} \cdots \text{QFT}^\dagger$ sandwich inside a Fourier-basis multiplier, which is where most of the gates are. **(b) Compensating residue:** in Theorem 39, only blocks (1), (5), (7) take the outer control (and (4) keeps its control on t); with the control off, blocks (2), (3), (4), (6) conspire to do nothing — $b - N$ is negative so t is set, block (4) adds N back, and block (6) clears t — but *only because* $b < N$. Both branches of the proof use that hypothesis: fed a value $\geq N$ the circuit silently leaves t dirty and the value register wrong by N , destroying the interference downstream. It holds here because every adder reduces mod N .

Finally, a doubly-controlled block becomes singly-controlled once and for all: with a fresh ancilla d , $c\text{-}c\text{-}V = \text{AND}(c_1, c_2 \rightarrow d)^\dagger \cdot c\text{-}V_d \cdot \text{AND}(c_1, c_2 \rightarrow d)$ on the subspace $d = 0$, and the temporary-AND pair costs 4 T gates total [12] (uncomputation is by X -basis measurement and a classically conditioned CZ, hence Clifford) regardless of how many blocks V contains.

Measured, $N = 15$, $a = 7$, transpiled to $\{\text{CNOT}, U\}$: controlling every gate of the modular adder gives 2 634 CNOTs / depth 4 836; the 3-of-7 construction gives 206 / 354 — a factor 12.8. For the full $c\text{-}U_a$: 20 892 / 37 727 versus 1 748 / 2 957, a factor 12.0. Framework methods (`QuantumCircuit.control`, `controlled_by`, `Controlled`) honestly control every gate; use them as the *specification*, then hand-build and assert equality *on the admissible subspace* — comparing full unitaries reports failures that are not real, and an unchecked ancilla can hide one that is.

E.5 Further optimisations, and the semiclassical variant

The tower of SE is the textbook construction. Everything below improves on it, and it is worth separating the improvements that change *the circuits above* from those that change *what is computed* and leave the circuits alone.

Technique	Kind	What it changes
windowing [14, 15]	circuit	a table lookup replaces the controlled blocks; $\approx 25\times$ fewer additions at $n = 2048$
coset representation [7, 15]	circuit	removes the explicit reduction: Theorem 39 collapses to a plain adder
oblivious carry runways [15]	circuit	segments a long adder so pieces run in parallel; depth $\Theta(n) \rightarrow \Theta(n/k)$
temporary AND [12]	circuit	4 T per compute/uncompute pair instead of 7 each
residue arithmetic [18, 16]	circuit	replaces the <i>representation</i> of a residue; $n/2 + o(n)$ qubits asymptotically
semiclassical QFT [4]	circuit	the counting register becomes one recycled qubit
Ekerå-Håstad [17]	algorithmic	shortens the <i>ladder</i> ($2n \rightarrow 1.5n$ rungs); the rungs are unchanged
Regev [20]	algorithmic	$\prod_i a_i^{z_i}$ in place of a^x : fewer multiplications per run, more runs and more space
Fibonacci exponentiation [21]	circuit	implements Regev's product in $O(n \log n)$ qubits; needs a quantum-quantum multiplier

Windowing [14], as instantiated for factoring in [15]. Fix an exponent window c_e and a multiplication window c_m and precompute classically the table of the needed multiples of powers of a : the modular exponentiation then needs $\frac{2n}{c_e} \cdot \frac{n}{c_m}$ modular additions instead of $2n \cdot n$, each preceded by one $2^{c_e+c_m}$ -entry lookup. A lookup by unary iteration costs $2^{c_e+c_m} - 1$ Toffoli against $\approx n$ per addition, so the windows must satisfy $c_e + c_m \lesssim \log_2 n$; at $n = 2048$ that permits windows of about 5 and 5, cutting the addition count by $\approx 25\times$ [15]. Conditioning is thereby absorbed into data: the exponent bits *address* the table, so no separate controlled- U block appears in

modern constructions. The controlled constant adder of §E, in which only the loading of the constant carries the control, is the $c_e = c_m = 1$ corner of exactly this idea.

Coset representation of modular integers [7], used throughout [15]. Pad the value register with a few extra qubits and represent $x \bmod N$ not by its least residue but by a superposition of coset representatives $x, x + N, x + 2N, \dots$. Adding a constant to a coset representative gives another coset representative, so **the reduction step disappears**: the seven blocks of Theorem 39 collapse to one plain addition. The price is an approximation — the padding bounds the probability that a representative walks off the top of the register — and a few extra qubits. Of all the savings listed here this is the one that most changes the circuits of §E, and it also retires the hypothesis $b < N$ that Theorem 39 and Lemma 41 both lean on.

Oblivious carry runways [15]. The carry chain, not the gate count, is what makes an n -bit addition $\Theta(n)$ deep. Insert extra “runway” qubits at intervals: a carry reaching a runway is absorbed there rather than propagating further, so the register splits into segments that may be added in parallel, at depth $\Theta(n/k)$ for k runways. Again the saving is bought with an approximation, the runways being merged back only occasionally.

Residue arithmetic [18], sharpened in [16]. Chevignard, Fouque and Schrottenloher obtain the least significant bits of the modular exponentiation using only $o(\log N)$ work qubits, by carrying intermediate values in a *residue number system* — as a tuple of remainders modulo small coprime moduli — combined with the truncation of May and Schlieper [8] and the Ekerå–Håstad variant. This is not an optimisation of the tower in §E but a replacement for its representation of a residue: with no single register holding $x \bmod N$, there is nothing for a modular adder to reduce. Concretely 1730 logical qubits and 2^{36} Toffoli per run over ≈ 40 runs for RSA-2048. Gidney [16] makes the residue arithmetic *approximate* and cuts the Toffoli count by more than $100\times$; that, with the error-correction advances of §G, is what takes the physical-qubit estimate below 10^6 .

Fibonacci exponentiation [21]. Regev’s algorithm [20] is cheap in gates per run but expensive in space, needing $O(n^{3/2})$ qubits to hold its many registers. Ragavan and Vaikuntanathan replace binary exponentiation by a recurrence indexed by Fibonacci numbers — adapting Kaliski’s classical reversible exponentiation — so that registers can be reused, reaching $O(n \log n)$ qubits at $O(n^{3/2} \log n)$ gates: the gate count of Regev with the space of Shor. Note that their inner primitive is an in-place *quantum–quantum* modular multiplication, a different block from the quantum-times-classical-constant multiplier of §E, in which one operand is known at compile time.

One control qubit instead of t . Every counting qubit is measured immediately after its rotations in QFT[†] and never reused, so each such controlled rotation may be replaced by a classically conditioned one-qubit rotation applied after measuring the control (the Griffiths–Niu semiclassical QFT [4]). The whole counting register collapses to a single recycled qubit, t rounds of measure-and-feed-forward, at no gate cost. Beauregard’s $2n + 3$ -qubit circuit [10] depends on this.

Two lineages. The line this document sits on is Shor’s structure with progressively better arithmetic: windowing, then coset representation and carry runways [15], then residue arithmetic [18, 16]. The other line changes the algorithm: Regev [20], made practical in space by Ragavan–Vaikuntanathan [21]. Ekerå–Håstad [17] is orthogonal to both and composes with either.

What none of them changes. Every construction above still spends essentially all of its gates on reversible modular arithmetic, and still reads the answer with a Fourier transform costing a negligible fraction of the total. The 2026 architecture results (§G) reduce the physical cost by two further orders of magnitude without touching the arithmetic at all: their Toffoli budgets are still built from ripple-carry adders, carry-lookahead adders and unary lookup tables.

Appendix F Proof that a random base works

This discharges Theorem 2.

Proof. By CRT, $(\mathbb{Z}/N\mathbb{Z})^\times \cong \prod_j (\mathbb{Z}/N_j\mathbb{Z})^\times$ with $N_j = p_j^{\alpha_j}$, and drawing a uniformly is drawing the components a_j independently and uniformly. Let $r_j = \text{ord}_{N_j}(a_j)$, so $r = \text{lcm}(r_j)$; write v_2 for the 2-adic valuation, $d_j := v_2(r_j)$, $d := v_2(r) = \max_j d_j$.

Step 1: the bad event forces all d_j equal. If r is odd, every $r_j \mid r$ is odd, so all $d_j = 0$. If r is even and $a^{r/2} \equiv -1$, then reducing mod N_j gives $a_j^{r/2} \equiv -1 \not\equiv 1$ (as N_j is odd) while $a_j^{r_j} \equiv 1$; hence $r_j \mid r$ but $r_j \nmid r/2$. Were $d_j < d$, the odd part of r_j would divide the odd part of $r/2$ (equal to that of r) and $v_2(r_j) \leq d - 1 = v_2(r/2)$, so $r_j \mid r/2$ — contradiction. So $d_j = d$ for all j .

Step 2: each d_j is spread out. For odd p , $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ is cyclic of even order $n_j = \varphi(N_j)$; write $c_j = v_2(n_j) \geq 1$ and $a_j = g_j^{k_j}$ with k_j uniform mod n_j . Then $r_j = n_j / \gcd(n_j, k_j)$, so $d_j = c_j - \min(v_2(k_j), c_j)$; since $\Pr[v_2(k_j) = e] = 2^{-(e+1)}$ for $0 \leq e < c_j$, every value has $\Pr[d_j = d] \leq \frac{1}{2}$.

Step 3. Fix j_0 . By independence, $\Pr[d_1 = \dots = d_m] = \sum_{d \geq 0} \prod_j \Pr[d_j = d] \leq \sum_{d \geq 0} \Pr[d_{j_0} = d] \cdot 2^{-(m-1)} = 2^{-(m-1)}$. $\square$

Appendix G Resource estimates in detail

Figures below last checked August 2026; they are date-sensitive and should be rechecked before use.

Concrete estimates for RSA-2048. Two figures circulate for the same problem; they optimise *different* things, and are often conflated.

	Beauregard [10]	Gidney–Ekerå [15]	Gidney 2025 [16]
arithmetic	Fourier (Draper) adder	ripple-carry, windowed	approximate residue arithmetic
logical qubits	$2n + 3$	a bit more than $3n$ (≈ 6200)	≈ 1400 , i.e. $\approx 0.7n$
gate count	$O(n^3 \log n)$ elem.*	$\approx 0.3n^3 \approx 3 \times 10^9$ Tof	6.5×10^9 Tof/run, ≈ 9 runs
physical qubits	—	20 million	$\approx 9 \times 10^5$ (rounded to 10^6)
runtime	—	8 hours	≈ 5 days (rounded to a week)
key technique	QFT adder	temporary AND; windowing	yoked surface codes; magic-state cultivation

*Beauregard’s cost is quoted in *elementary gates*: his Fourier adder is built from controlled rotations, each costing $\Theta(\log(1/\epsilon))$ T gates fault-tolerantly, so that column is not comparable Toffoli-for-Toffoli with the other two. A gate count is interpretable only alongside the construction it came from. The minimal-*width* circuit of Häner–Roetteler–Svore [11] uses only $2n + 2 = 4098$ logical qubits but pays $(64(\log_2 n - 2) + 29.46)n^3 \approx 5.2 \times 10^{12}$ Toffoli — a spread of $\sim 10^3$ in gates against Gidney–Ekerå at *identical* asymptotics; Chevnard–Fouque–Schrottenloher [18] reach $n/2 + o(n)$ logical qubits *asymptotically*, but their concrete figure for RSA-2048 is 1730 logical qubits, i.e. $\approx 0.85n$, at 2^{36} Toffoli per run over ≈ 40 runs ($\approx 2 \times 10^{12}$ Toffoli in total), which Gidney 2025 cuts by $> 100\times$. Quoting the asymptotic alongside concrete counts understates the width by about $1.7\times$. The physical estimates assume a uniform 10^{-3} gate error rate (below the surface-code threshold), a $1 \mu\text{s}$ cycle time, a $10 \mu\text{s}$ reaction time and fast decoding. They give the hardware requirement at that quality level; the date at which such hardware exists is a separate question.

Variants that move the numbers. Ekerå–Håstad [17] exploits that an RSA modulus has *balanced* prime factors, so factoring reduces to a *short* discrete logarithm: in the single-run variant used by Gidney–Ekerå the exponent register shrinks from $2n$ to $n_e = 1.5n + O(1)$ qubits — a $4/3$ reduction in modular multiplications — and a tradeoff parameter drives n_e towards $\approx n/2$ at the price of several runs and a harder lattice-enumeration post-processing. It does not apply to a general ECDLP. Regev [20] replaces the one-dimensional structure by a d -dimensional one, giving $\tilde{O}(n^{3/2})$ gates per run at the price of $\approx \sqrt{n} + 4$ runs and lattice reduction; the total is still $\tilde{O}(n^2)$ gates, the same as Shor with fast multiplication, so the gain is in per-run circuit *depth*, the quantity constrained by coherence budgets. Ragavan–Vaikuntanathan [21] brings its qubit count back to $O(n \log n)$.

Appendix H Two further readings of the inverse QFT

§3.3 gave the matched-filter reading. Here are two others, each useful in a different context.

(1) *As the inverse of a known encoding.* By the product form (26), $\text{QFT}_Q|b\rangle = Q^{-1/2} \sum_x e^{2\pi i x b/Q} |x\rangle$: the Fourier transform of the *position* b is a phase *ramp* of slope b/Q . The kickback ladder builds exactly such a ramp, of slope ϕ . So if $\phi = b/Q$ the ladder has literally prepared $\text{QFT}_Q|b\rangle$ — the counting register is holding the number b , but written in the Fourier basis, where it is invisible to a computational-basis measurement because every amplitude has modulus $Q^{-1/2}$. Applying $\text{QFT}_Q^\dagger$ is nothing but a change of coordinates back to the basis the measurement can read, and it returns $|b\rangle$ exactly (Theorem 10). The information was moved from the phases into the labels.

(2) *As the duality of Theorem 26, run backwards.* Write $D_\phi := \text{diag}(e^{2\pi i x \phi})_x$ for the modulation the ladder applies. Then the state before the transform is $D_\phi \text{QFT}_Q|0\rangle$, and (20) read as $S^b = F^\dagger D^b F$ says: *modulation in one basis is translation in the other*. A phase gradient of slope b/Q is, viewed through $F^\dagger$, a translation of the pointer $|0\rangle$ by b places. The inverse QFT converts a *rate of winding* into a *displacement*, and the measurement reads the displacement off as an integer. This is why the same matrix F appears in §B as the diagonaliser of the shift and here as the read-out device: the read-out applies the same diagonalisation in reverse.

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